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CMOS Processing Technology

3

3.1 Introduction

Chapter 1 summarized the steps in a basic CMOS process. These steps are expanded upon in this chapter. Where possible, the processing details are related to the way CMOS circuits and systems are designed. Modern CMOS processing is complex, and while coverage of every nuance is not within the scope of this book, we focus on the fundamental concepts that impact design.

A fair question from a designer would be "Why do I care how transistors are made?" In many cases, if designers understand the physical process, they will comprehend the reason for the underlying design rules and in turn use this knowledge to create a better design. Understanding the manufacturing steps is also important when debugging some difficult chip failures and improving yield.

Fabrication plants, or *fabs*, are enormously expensive to develop and operate. In the early days of the semiconductor industry, a few bright physicists and engineers could bring up a fabrication facility in an industrial building at a modest cost and most companies did their own manufacturing. Now, a fab processing 300 mm wafers in a 45 nm process costs about $3 billion. The research and development underlying the technology costs another $2.4 billion. Only a handful of companies in the world have the sales volumes to justify such a large investment. Even these companies are forming consortia to share the costs of technology development with their market rivals. Some companies, such as TSMC, UMC, Chartered, and IBM operate on a *foundry* model, selling space on their fab line to fabless semiconductor firms. Figure 3.1 shows workers and machinery in the cavernous clean room at IBM's East Fishkill 300 mm fab.

Recall that silicon in its pure or *intrinsic* state is a semiconductor, having bulk electrical resistance somewhere between that of a conductor and an insulator. The conductivity of silicon can be raised by several orders of magnitude by introducing *impurity* atoms into the silicon crystal lattice. These dopants can supply either free electrons or holes. Group III impurity elements such as boron that use up electrons are referred to as *acceptors* because they accept some of the electrons already in the silicon, leaving holes. Similarly, Group V *donor* elements such as arsenic and phosphorous provide electrons. Silicon that contains a majority of donors is known as *n-type*, while silicon that contains a majority of acceptors is known as *p-type*. When n-type and p-type materials are brought together, the region where the silicon changes from n-type to p-type is called a *junction*. By arranging junctions in certain physical structures and combining them with wires and insulators, various semiconductor devices can be constructed. Over the years, silicon semiconductor processing has evolved sophisticated techniques for building these junctions and other insulating and conducting structures.



FIGURE 3.1 IBM, East Fishkill, NY fab (Courtesy of International Business Machines Corporation. Unauthorized use not permitted.)

The chapter begins with the steps of a generic process characteristic of commercial 65 nm manufacturing. It also surveys a variety of process enhancements that benefit certain applications. The chapter examines layout design rules in more detail and discusses layout CAD issues such as design rule checking.

3.2 CMOS Technologies

CMOS processing steps can be broadly divided into two parts. Transistors are formed in the *Front-End-of-Line* (FEOL) phase, while wires are built in the *Back-End-of-Line* (BEOL) phase. This section examines the steps used through both phases of the manufacturing process.

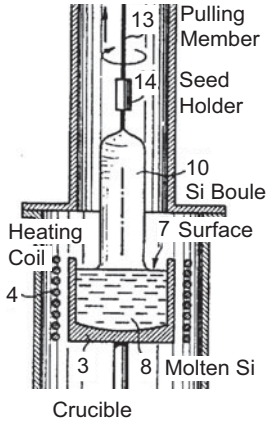


FIGURE 3.2 Czochralski system for growing Si boules (Adapted from [Schulmann98].)

3.2.1 Wafer Formation

The basic raw material used in CMOS fabs is a *wafer* or disk of silicon, roughly 75 mm to 300 mm (12″—a dinner plate!) in diameter and less than 1 mm thick. Wafers are cut from *boules*, cylindrical ingots of single-crystal silicon, that have been pulled from a crucible of pure molten silicon. This is known as the *Czochralski* method and is currently the most common method for producing single-crystal material. Controlled amounts of impurities are added to the melt to provide the crystal with the required electrical properties. A seed crystal is dipped into the melt to initiate crystal growth. The silicon ingot takes on the same crystal orientation as the seed. A graphite radiator heated by radio-frequency induction surrounds the quartz crucible and maintains the temperature a few degrees above the melting point of silicon (1425 °C). The atmosphere is typically helium or argon to prevent the silicon from oxidizing.

The seed is gradually withdrawn vertically from the melt while simultaneously being rotated, as shown in Figure 3.2. The molten silicon attaches itself to the seed and recrystallizes as it is withdrawn. The seed withdrawal and rotation rates determine the diameter of the ingot. Growth rates vary from 30 to 180 mm/hour.

3.2.2 Photolithography

Recall that regions of dopants, polysilicon, metal, and contacts are defined using masks. For instance, in places covered by the mask, ion implantation might not occur or the dielectric or metal layer might be left intact. In areas where the mask is absent, the implantation can occur, or dielectric or metal could be etched away. The patterning is achieved by a process called *photolithography*, from the Greek *photo* (light), *lithos* (stone), and *graphe* (picture), which literally means "carving pictures in stone using light." The primary method for defining areas of interest (i.e., where we want material to be present or absent) on a wafer is by the use of *photoresists*. The wafer is coated with the photoresist and subjected to selective illumination through the *photomask*. After the initial patterning of photoresist, other barrier layers such as polycrystalline silicon, silicon dioxide, or silicon nitride can be used as physical masks on the chip. This distinction will become more apparent as this chapter progresses.

A photomask is constructed with chromium (chrome) covered quartz glass. A UV light source is used to expose the photoresist. Figure 3.3 illustrates the lithography process. The photomask has chrome where light should be blocked. The UV light floods the mask from the backside and passes through the clear sections of the mask to expose the organic photoresist (PR) that has been coated on the wafer. A developer solvent is then used to dissolve the soluble unexposed photoresist, leaving islands of insoluble exposed photoresist. This is termed a *negative* photoresist. A *positive* resist is initially insoluble, and when exposed to UV becomes soluble. Positive resists provide for higher resolution than negative resists, but are less sensitive to light. As feature sizes become smaller, the photoresist layers have to be made thinner. In turn, this makes them less robust and more subject to failure which can impact the overall yield of a process and the cost to produce the chip.

The photomask is commonly called a *reticle* and is usually smaller than the wafer, e.g., 2 cm on a side. A *stepper* moves the reticle to successive locations to completely expose the wafer. *Projection printing* is normally used, in which lenses between the reticle and wafer focus the pattern on the wafer surface. Older techniques include *contact printing*, where the mask and wafer are in contact, and *proximity printing*, where the mask and wafer are close but not touching. The reticle can be the same size as the area to be patterned (1×) or larger. For instance, 2.5× and 5× steppers with optical reduction have been used in the industry.

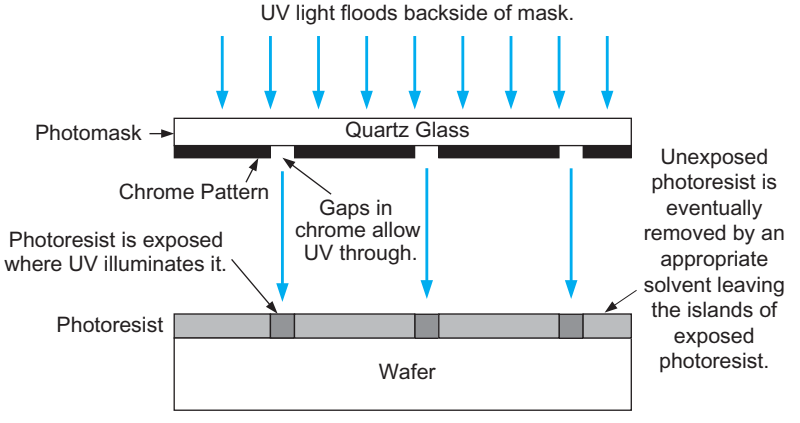


FIGURE 3.3 Photomasking with a negative resist (lens system between mask and wafer omitted to improve clarity and avoid diffracting the reader ☺)

The wavelength of the light source influences the minimum feature size that can be printed. Define the minimum *pitch* (width + spacing) of a process to be $2b$. The resolution of a lens depends on the wavelength λ of the light and the *numerical aperture* NA of the lens:

$$2b = k_1 \frac{\lambda}{\text{NA}} \tag{3.1}$$

The numerical aperture is

$$\text{NA} = n \sin \alpha \tag{3.2}$$

where n is the refractive index of the medium (1 for air, 1.33 for water, and up to 1.5 for oil), and α is the angle of acceptance of the lens. Increasing α requires larger optics. Lenses used in the 1970s had a numerical aperture of 0.2. Intel uses a numerical aperture of 0.92 for their 45 nm process [Mistry07]. Nikon and ASML broke the 1.0 barrier by introducing *immersion lithography* that takes advantage of water's higher refractive index [Geppert04], and in 2008, NA = 1.35 had been reached. All of these advances have come at the expense of multimillion dollar optics systems. k_1 depends on the coherence of the light, antireflective coatings, photoresist parameters, and resolution enhancement techniques. Presently, 0.8 is considered easy, while 0.5 is very hard.

The *depth of focus* is

$$\text{DOF} = \frac{k_2 \lambda}{\text{NA}^2} \tag{3.3}$$

where k_2 ranges from 0.5 to 1. Advanced lithography systems with short wavelengths and large numerical apertures have a very shallow depth of focus, requiring that the surface of the wafer be maintained extremely flat.

In the 1980s, mercury lamps with 436 nm or 365 nm wavelengths were used. At the 0.25 μm process generation, excimer lasers with 248 nm (deep ultraviolet) were adopted and have been used down to the 180 nm node. Currently, 193 nm argon-fluoride lasers are used for the *critical layers* down to the 45 nm node and beyond. The critical layers are those that define the device behavior. An example would be the gate (polysilicon), source/drain (diffusion), first metal, and contact masks. With such a laser, a numerical aperture of 1.35, and $k_1 = 0.5$, the best achievable pitch is $2b = 72$ nm, corresponding to a polysilicon half-pitch of 36 nm. It is amazing that we can print features so much smaller than the wavelength of the light, but even so, lithography is becoming a serious problem at the 45 nm node and below.

Efforts to develop 157 nm deep UV lithography systems were unsuccessful and have been abandoned by the industry. In the future, 13.5 nm extreme ultraviolet (EUV) light sources may be used, but presently, these sources require prohibitively expensive reflective optics and vacuum processing and are not strong enough for production purposes. Some predict that EUV will be ready by 2011 or 2012, while others are skeptical [Mack08].

Wavelengths comparable to or greater than the feature size cause distortion in the patterns exposed on the photoresist. *Resolution enhancement techniques* (RETs) precompensate for this distortion so the desired patterns are obtained [Schellenberg03]. These techniques involve modifying the amplitude, phase, or direction of the incoming light. The

ends of a line in a layout receive less light than the center, causing nonuniform exposure. *Optical proximity correction* (OPC) makes small changes to the patterns on the masks to compensate for these local distortions. Figure 3.4 shows an example of printing with and without optical proximity correction. OPC predistorts the corners to reduce undesired rounding. *Phase shift masks* (PSM) takes advantage of the diffraction grating effect of parallel lines on a mask, varying the thickness of the mask to change the phase such that light from adjacent lines are out of phase and cancel where no light is desired. *Off-axis illumination* can also improve contrast for certain types of dense, repetitive patterns. *Double-patterning* is a sequence of two precisely aligned exposure steps with different masks for the same photoresist layer [Mack08]. OPC became necessary at the 180 nm node and all of these techniques are in heavy use by the 45 nm node.

Each successive UV stepper is more expensive and the throughput of the stepper may decrease. This is just another contributory issue to the spiraling cost of chip manufacturing. The cost of masks is also skyrocketing, forcing chip designers to amortize design and mask expenses across the largest volume possible. This theme will be reinforced in Section 14.3.

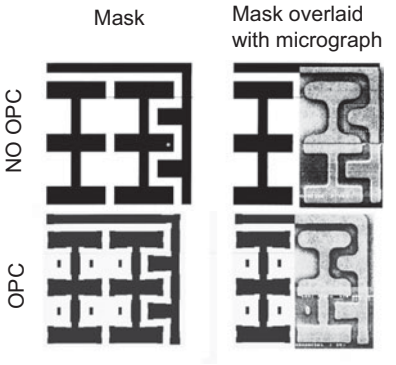


FIGURE 3.4 Subwavelength features printed with and without OPC. Predistortion of corners in OPC reduces undesired rounding. (Adapted from [Schellenberg98] with permission of SPIE.)

3.2.3 Well and Channel Formation

The following are main CMOS technologies:

- n-well process
- p-well process
- twin-well process
- triple-well process

Silicon-on-insulator processes are also available through some manufacturers (see Section 3.4.1.2).

Chapter 1 outlined an n-well process. Historically, p-well processes preceded n-well processes. In a p-well process, the nMOS transistors are built in a p-well and the pMOS transistor is placed in the n-type substrate. p-well processes were used to optimize the pMOS transistor performance. Improved techniques allowed good pMOS transistors to be fabricated in an n-well and excellent nMOS transistors to be fabricated in the p-type substrate of an n-well process. In the n-well process, each group of pMOS transistors in an n-well shares the same body node but is isolated from the bodies of pMOS transistors in different wells. However, all the nMOS transistors on the chip share the same body, which is the substrate. Noise injected into the substrate by digital circuits can disturb sensitive analog or memory circuits. Twin-well processes accompanied the emergence of n-well processes. A twin-well process allows the optimization of each transistor type. A third well can be added to create a triple-well process. The triple-well process has emerged to provide good isolation between analog and digital blocks in mixed-signal chips; it is also used to isolate high-density dynamic memory from logic. Most fabrication lines provide a baseline twin-well process that can be upgraded to a triple-well process with the addition of a single mask level.

Wells and other features require regions of doped silicon. Varying proportions of donor and acceptor dopants can be achieved using *epitaxy*, *deposition*, or *implantation*.

Epitaxy involves growing a single-crystal film on the silicon surface (which is already a single crystal) by subjecting the silicon wafer surface to an elevated temperature and a source of dopant material.

Epitaxy can be used to produce a layer of silicon with fewer defects than the native wafer surface and also can help prevent latchup (see Section 7.3.6). Foundries may provide a choice of *epi* (with epitaxial layer) or non-epi wafers. Microprocessor designers usually prefer to use epi wafers for uniformity of device performance.

Deposition involves placing dopant material onto the silicon surface and then driving it into the bulk using a thermal diffusion step. This can be used to build deep junctions. A step called *chemical vapor deposition* (CVD) can be used for the deposition. As its name suggests, CVD occurs when heated gases react in the vicinity of the wafer and produce a product that is deposited on the silicon surface. CVD is also used to lay down thin films of material later in the CMOS process.

Ion implantation involves bombarding the silicon substrate with highly energized donor or acceptor atoms. When these atoms impinge on the silicon surface, they travel below the surface of the silicon, forming regions with varying doping concentrations. At elevated temperature (>800 °C) diffusion occurs between silicon regions having different densities of impurities, with impurities tending to diffuse from areas of high concentration to areas of low concentration. Therefore, it is important to keep the remaining process steps at as low a temperature as possible once the doped areas have been put into place. However, a high-temperature annealing step is often performed after ion implantation to redistribute dopants more uniformly. Ion implantation is the standard well and source/drain implant method used today. The placement of ions is a random process, so doping levels cannot be perfectly controlled, especially in tiny structures with relatively small numbers of dopant atoms. Statistical dopant fluctuations lead to variations in the threshold voltage that will be discussed in Section 7.5.2.2.

The first step in most CMOS processes is to define the well regions. In a triple-well process, a deep n-well is first driven into the p-type substrate, usually using high-energy Mega electron volt levels (MeV) ion implantation as opposed to a thermally diffused operation. This avoids the thermal cycling (i.e., the wafers do not have to be raised significantly in temperature), which improves throughput and reliability. A 2–3 MeV implantation can yield a 2.5–3.5 μm deep n-well. Such a well has a peak dopant concentration just under the surface and for this reason is called a *retrograde* well. This can enhance device performance by providing improved latchup characteristics and reduced susceptibility to vertical punch-through (see Section 7.3.5). A thick (3.5–5.5 μm) resist has to be used to block the high energy implantation where no well should be formed. Thick resists and deep implants necessarily lead to fairly coarse feature dimensions for wells, compared to the minimum feature size. Shallower n-well and p-well regions are then implanted. After the wells have been formed, the doping levels can be adjusted (using a *threshold implant*) to set the desired threshold voltages for both nMOS and pMOS transistors. With multiple threshold implant masks, multiple V_t options can be provided on the same chip. For a given gate and substrate material, the threshold voltage depends on the doping level in the substrate (N_A), the oxide thickness (t_{ox}), and the surface state charge (Q_{fc}). The implant can affect both N_A and Q_{fc} and hence V_t. Figure 3.5 shows a typical triple-well structure. As discussed, the nMOS transistor is situated in the p-well located in the deep n-well. Other nMOS transistors could be built in different p-wells so that they do not share the same body node. Transistors in a p-well in a triple-well process will have different characteristics than transistors in the substrate because of the different doping levels. The pMOS

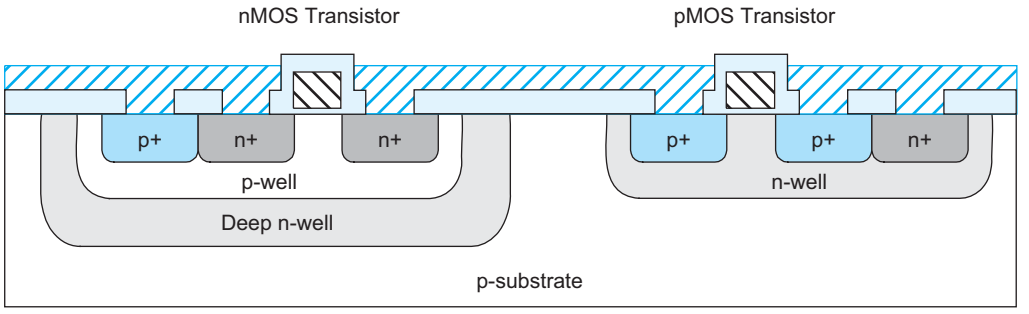


FIGURE 3.5 Well structure in triple-well process

transistors are located in the shallow (normal) n-well. The figure shows the cross-section of an inverter.

Wells are defined by separate masks. In the case of a twin-well process, only one mask need be defined because the other well by definition is its complement. Triple-well processes have to define at least two masks, one for the deep well and the other for either n-well or p-well.

Transistors near the edge of a retrograde well (e.g., within about 1 μm) may have different threshold voltages than those far from the edge because ions scatter off the photoresist mask into the edge of the well, as shown in Figure 3.6 [Hook03]. This is called the *well-edge proximity effect*.

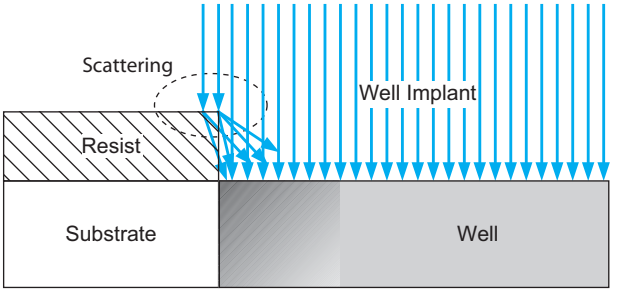


FIGURE 3.6 Well-edge proximity effect, in which dopants scattering off photoresist increase the doping level near the edge of a well (© IEEE 2003.)

3.2.4 Silicon Dioxide (SiO_2)

Many of the structures and manufacturing techniques used to make silicon integrated circuits rely on the properties of SiO_2. Therefore, reliable manufacture of SiO_2 is extremely important. In fact, unlike competing materials, silicon has dominated the industry because it has an easily processable oxide (i.e., it can be grown and etched). Various thicknesses of SiO_2 may be required, depending on the particular process. Thin oxides are required for transistor gates; thicker oxides might be required for higher voltage devices, while even thicker oxide layers might be required to ensure that transistors are not formed unintentionally in the silicon beneath polysilicon wires (see the next section).

Oxidation of silicon is achieved by heating silicon wafers in an oxidizing atmosphere. The following are some common approaches:

- *Wet oxidation*—when the oxidizing atmosphere contains water vapor. The temperature is usually between 900 °C and 1000 °C. This is also called *pyrogenic* oxidation when a 2:1 mixture of hydrogen and oxygen is used. Wet oxidation is a rapid process.
- *Dry oxidation*—when the oxidizing atmosphere is pure oxygen. Temperatures are in the region of 1200 °C to achieve an acceptable growth rate. Dry oxidation forms a better quality oxide than wet oxidation. It is used to form thin, highly controlled gate oxides, while wet oxidation may be used to form thick field oxides.
- *Atomic layer deposition (ALD)*—when a thin chemical layer (material *A*) is attached to a surface and then a chemical (material *B*) is introduced to produce a thin layer of the required layer (i.e., SiO_2—this can also be used for other various dielectrics and metals). The process is then repeated and the required layer is built up layer by layer. [George96, Klaus98].

The oxidation process normally consumes part of the silicon wafer (deposition and ALD do not). Since SiO_2 has approximately twice the volume of silicon, the SiO_2 layer grows almost equally in both vertical directions. Thus, after processing, the SiO_2 projects above and below the original unoxidized silicon surface.

3.2.5 Isolation

Individual devices in a CMOS process need to be isolated from one another so that they do not have unexpected interactions. In particular, channels should only be inverted beneath transistor gates over the active area; wires running elsewhere shouldn't create parasitic MOS channels. Moreover, the source/drain diffusions of unrelated transistors should not interfere with each other.

The process flow in Section 1.5 was historically used to provide this isolation. The transistor gate consists of a thin *gate oxide* layer. Elsewhere, a thicker layer of field oxide separates polysilicon and metal wires from the substrate. The MOS sandwich formed by the wire, thick oxide, and substrate behaves as an unwanted parasitic transistor. However, the thick oxide effectively sets a threshold voltage greater than V_{DD} that prevents the transistor from turning ON during normal operation. Actually, these *field devices* can be used for I/O protection and are discussed in Section 13.6.2. The source and drain of the transistors form reverse-biased p–n junctions with the substrate or well, isolating them from their neighbors.

The thick oxide used to be formed by a process called *Local Oxidation of Silicon* (LOCOS). A problem with LOCOS-based processes is the transition between thick and thin oxide, which extended some distance laterally to form a so-called *bird's beak*. The lateral distance is proportional to the oxide thickness, which limits the packing density of transistors.

Starting around the 0.35 μm node, *shallow trench isolation* (STI) was introduced to avoid the problems with LOCOS. STI forms insulating trenches of SiO_2 surrounding the transistors (everywhere except the active area). The trench width is independent of its depth, so transistors can be packed as closely as the lithography permits. The trenches isolate the wires from the substrate, preventing unwanted channel formation. They also reduce the sidewall capacitance and junction leakage current of the source and drain.

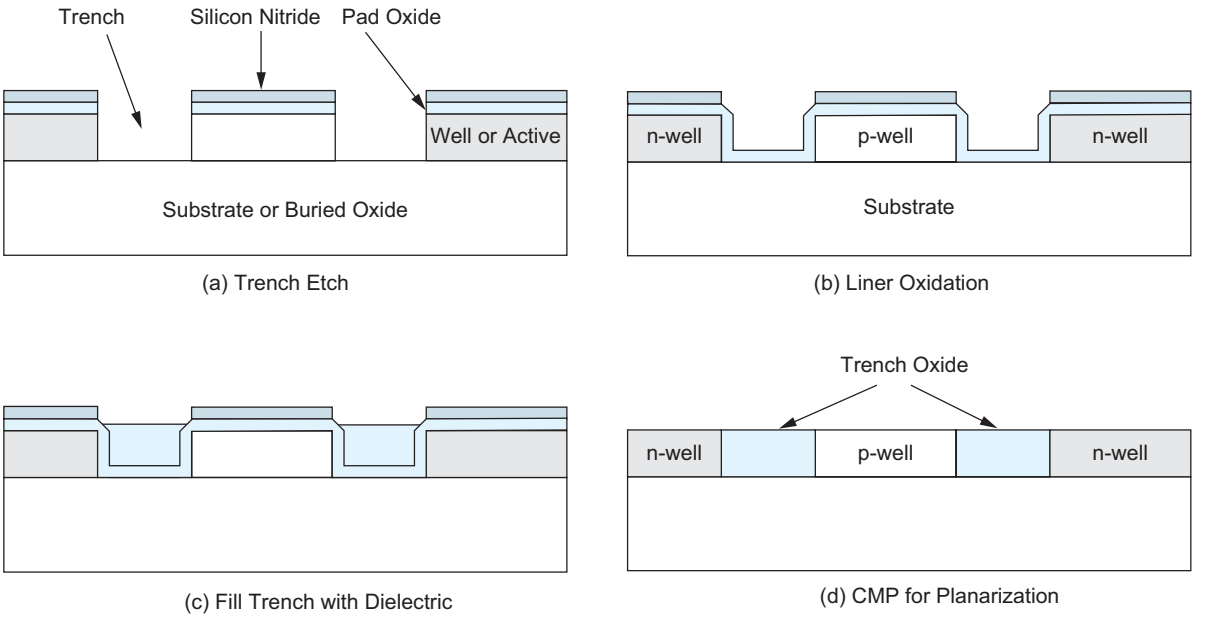


FIGURE 3.7 Shallow trench isolation

STI starts with a *pad oxide* and a silicon nitride layer, which act as the masking layers, as shown in Figure 3.7. Openings in the pad oxide are then used to etch into the well or substrate region (this process can also be used for source/drain diffusion). A liner oxide is then grown to cover the exposed silicon (Figure 3.7(b)). The trenches are filled with SiO_2 or other fillers using CVD that does not consume the underlying silicon (Figure 3.7(c)). The pad oxide and nitride are removed and a *Chemical Mechanical Polishing* (CMP) step is used to planarize the structure (Figure 3.7(d)). CMP, as its name suggests, combines a mechanical grinding action in which the rotating wafer is contacted by a stationary polishing head while an abrasive mixture is applied. The mixture also reacts chemically with the surface to aid in the polishing action. CMP is used to achieve flat surfaces, which are of central importance in modern processes with many layers.

From the designer's perspective, the presence of a deep n-well and/or trench isolation makes it easier to isolate noise-sensitive (analog or memory) portions of a chip from digital sections. Trench isolation also permits nMOS and pMOS transistors to be placed closer together because the isolation provides a higher source/drain *breakdown voltage*—the voltage at which a source or drain diode starts to conduct in the reverse-biased condition. The breakdown voltage must exceed the supply voltage (so junctions do not break down during normal operation) and is determined by the junction dimensions and doping levels of the junction formed. Deeper trenches increase the breakdown voltage.

3.2.6 Gate Oxide

The next step in the process is to form the gate oxide for the transistors. As mentioned, this is most commonly in the form of *silicon dioxide* (SiO_2).

In the case of STI-defined source/drain regions, the gate oxide is grown on top of the planarized structure that occurs at the stage shown in Figure 3.7(d). This is shown in

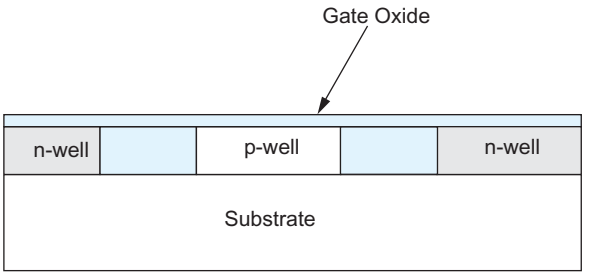


FIGURE 3.8 Gate oxide formation

Figure 3.8. The oxide structure is called the *gate stack*. This term arises because current processes seldom use a pure SiO_2 gate oxide, but prefer to produce a stack that consists of a few atomic layers, each 3–4 Å thick, of SiO_2 for reliability, overlaid with a few layers of an *oxynitrided* oxide (one with nitrogen added). The presence of the nitrogen increases the dielectric constant, which decreases the *effective oxide thickness* (EOT); this means that for a given oxide thickness, it performs like a thinner oxide. Being able to use a thicker oxide improves the robustness of the process. This concept is revisited in Section 3.4.1.3.

Many processes in the 180 nm generation and beyond provide at least two oxide thicknesses, as will be discussed in Section 3.4.1.1 (thin for logic transistors and thick for I/O transistors that must withstand higher voltages). At the 65 nm node, the effective thickness of the thin gate oxide is only 10.5–15 Å.

3.2.7 Gate and Source/Drain Formations

When silicon is deposited on SiO_2 or other surfaces without crystal orientation, it forms polycrystalline silicon, commonly called *polysilicon* or simply *poly*. An annealing process is used to control the size of the single crystal domains and to improve the quality of the polysilicon. Undoped polysilicon has high resistivity. The resistance can be reduced by implanting it with dopants and/or combining it with a refractory metal. The polysilicon gate serves as a mask to allow precise alignment of the source and drain on either side of the gate. This process is called a *self-aligned polysilicon gate* process. Aluminum could not be used because it would melt during formation of the source and drain.

As a historical note, early metal-gate processes first diffused source and drain regions, and then formed a metal gate. If the gate was misaligned, it could fail to cover the entire channel and lead to a transistor that never turned ON. To prevent this, the metal gate had to overhang the source and drain by more than the alignment tolerance of the process. This created large parasitic gate-to-source and gate-to-drain overlap capacitances that degraded switching speeds.

The steps to define the gate, source, and drain in a self-aligned polysilicon gate are as follows:

- Grow gate oxide wherever transistors are required (area = source + drain + gate)—elsewhere there will be thick oxide or trench isolation (Figure 3.9(a))
- Deposit polysilicon on chip (Figure 3.9(b))
- Pattern polysilicon (both gates and interconnect) (Figure 3.9(c))
- Etch exposed gate oxide—i.e., the area of gate oxide where transistors are required that was not covered by polysilicon; at this stage, the chip has windows down to the well or substrate wherever a source/drain diffusion is required (Figure 3.9(d))
- Implant pMOS and nMOS source/drain regions (Figure 3.9(e))

The source/drain implant density is relatively low, typically in the range 10^{18}–10^{20} cm^{-3} of impurity atoms. Such a *lightly doped drain* (LDD) structure reduces the electric field at the drain junction (the junction with the highest voltage), which improves the immunity of the device to hot electron damage (see Section 7.3.6) and suppresses short-channel effects. The LDD implants are shallow and lightly doped, so they exhibit low

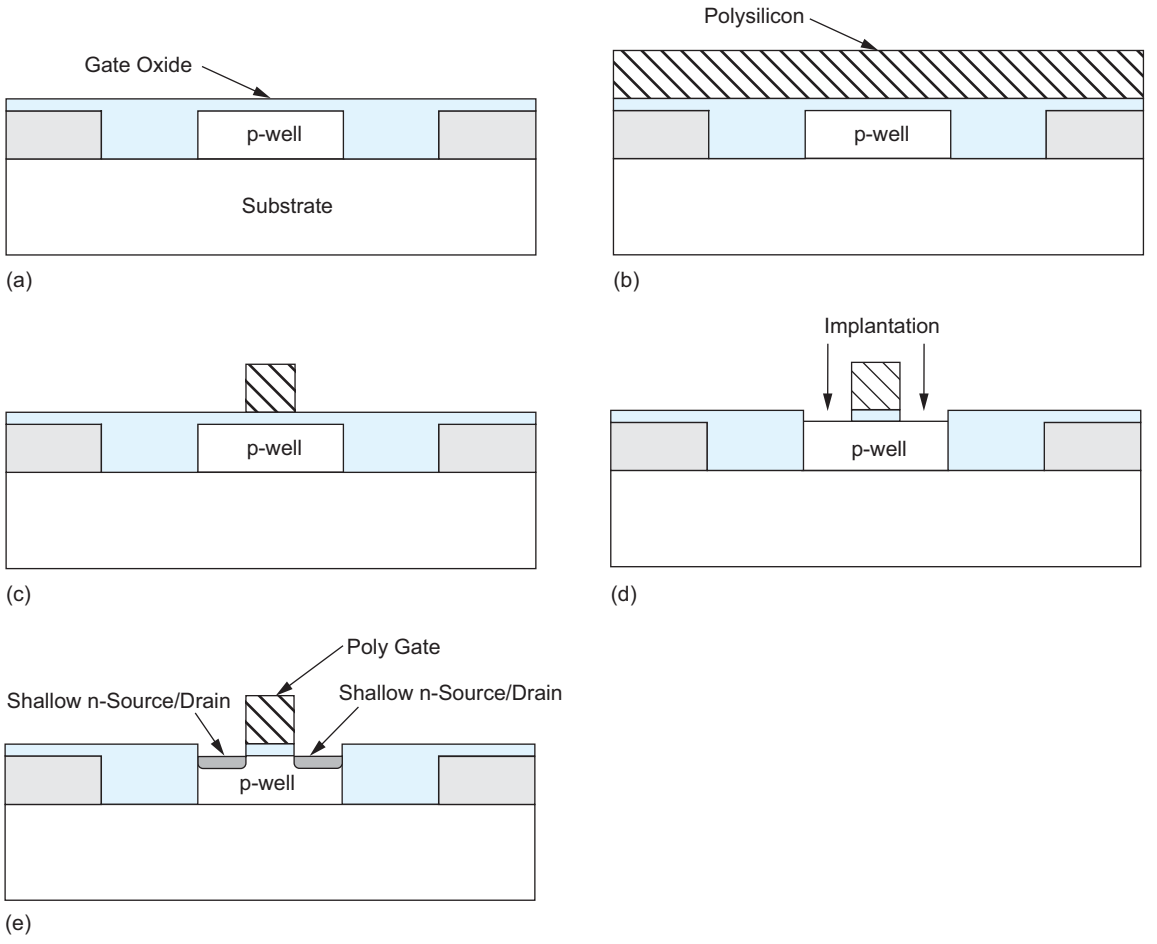


FIGURE 3.9 Gate and shallow source/drain definition

capacitance but high resistance. This reduces device performance somewhat because of the resistance in series with the transistor. Consequently, deeper, more heavily doped source/drain implants are needed in conjunction with the LDD implants to provide devices that combine hot electron suppression with low source/drain resistance. A silicon nitride (Si_3N_4) *spacer* along the edge of the gate serves as a mask to define the deeper diffusion regions, as shown in Figure 3.10(a). For in-depth coverage of various LDD structures, see [Ziegler02].

As mentioned, the polysilicon gate and source/drain diffusion have high resistance due to the resistivity of silicon and their extremely small dimensions. Modern processes form a surface layer of a refractory metal on the silicon to reduce the resistance. A *refractory metal* is one with a high melting point that will not be damaged during subsequent processing. Tantalum, nickel, molybdenum, titanium, or cobalt are commonly used. The metal is deposited on the silicon (specifically on the gate polysilicon and/or source/drain regions). A layer of *silicide* is formed when the two substances react at elevated temperatures. In a *polycide* process, only the gate polysilicon is silicided. In a silicide process (usually implemented as a self-aligned silicidization—from whence comes the synonymous

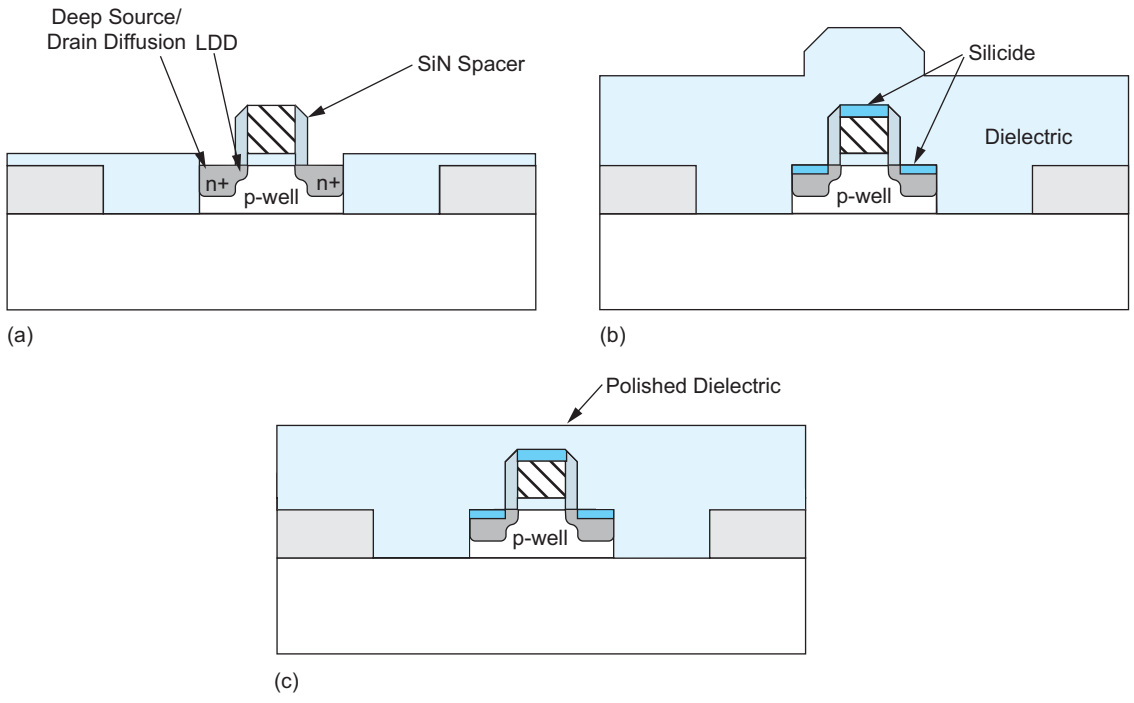


FIGURE 3.10 Transistor with LDD and deep diffusion, salicide, and planarized dielectric

term *salicide*) both gate polysilicon and source/drain regions are silicided. This process lowers the resistance of the polysilicon interconnect and the source and drain diffusion.

Figure 3.10(b) shows the resultant structure with gate and source/drain regions silicided. In addition, SiO_2 or an alternative dielectric has been used to cover all areas prior to the next processing steps. The figure shows a resulting structure with some vertical topology typical of older processes. The rapid transitions in surface height can lead to breaks in subsequent layers that fail to conform, or can entail a plethora of design rules that relate to metal edges. To avoid these problems, a CMP step is used to planarize the dielectric, leaving a flat surface for metallization as shown in Figure 3.10(c).

Nanometer processes involve another implantation step called *halo doping* that increases the doping of the substrate or well near the ends of the channels. The halo doping alleviates DIBL, short channel effects, and punchthrough but increases GIDL and BTBT leakage at the junction between the diffusion and channel [Roy03].

3.2.8 Contacts and Metallization

Contact cuts are made to source, drain, and gate according to the contact mask. These are holes etched in the dielectric after the source/drain step discussed in the previous section. Older processes commonly use aluminum (Al) for wires, although newer ones offer copper (Cu) for lower resistance. Tungsten (W) can be used as a *plug* to fill the contact holes (to alleviate problems of aluminum not conforming to small contacts). In some processes, the tungsten can also be used as a local interconnect layer.

Metallization is the process of building wires to connect the devices. As mentioned previously, conventional metallization uses aluminum. Aluminum can be deposited either

by *evaporation* or *sputtering*. Evaporation is performed by passing a high electrical current through a thick aluminum wire in a vacuum chamber. Some of the aluminum atoms are vaporized and deposited on the wafer. An improved form of evaporation that suffers less from contamination focuses an electron beam at a container of aluminum to evaporate the metal. Sputtering is achieved by generating a gas plasma by ionizing an inert gas using an RF or DC electric field. The ions are focused on an aluminum target and the plasma dislodges metal atoms, which are then deposited on the wafer.

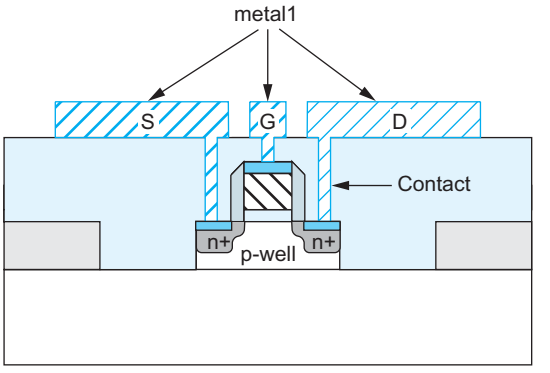


FIGURE 3.11 Aluminum metallization

Wet or *dry etching* can be used to remove unwanted metal. *Piranha solution* is a 3:1 to 5:1 mix of sulfuric acid and hydrogen peroxide that is used to clean wafers of organic and metal contaminants or photoresist after metal patterning. Plasma etching is a dry etch process with fluorine or chlorine gas used for metallization steps. The plasma charges the etch gas ions, which are attracted to the appropriately charged silicon surface. Very sharp etch profiles can be achieved using plasma etching. The result of the contact and metallization patterning steps is shown in Figure 3.11.

Subsequent intermetal vias and metallization are then applied. Some processes offer uniform metal dimensions for levels 2 to n–1, where n is the top level of metal. The top level is normally a thicker layer for use in power distribution and as such has relaxed width and spacing constraints. Other processes use successively thicker and wider metal for the upper layers, as will be explored in Section 6.1.2.

Polysilicon over diffusion normally forms a transistor gate, so a short metal1 wire is necessary to connect a diffusion output node to a polysilicon input. Some processes add tungsten (W) layer above polysilicon and below metal1; this layer is called *local interconnect* and can be drawn on a finer pitch than metal1. Local interconnect offers denser cell layouts, especially in static RAMs. Figure 3.12 shows a scanning electron micrograph of a partially completed SRAM array. The oxide has been removed to show the diffusion, polysilicon, local interconnect, and metal1. Local interconnect is used to connect the nMOS and pMOS transistors without rising up to metal1. SRAM cells are discussed further in Section 12.2.

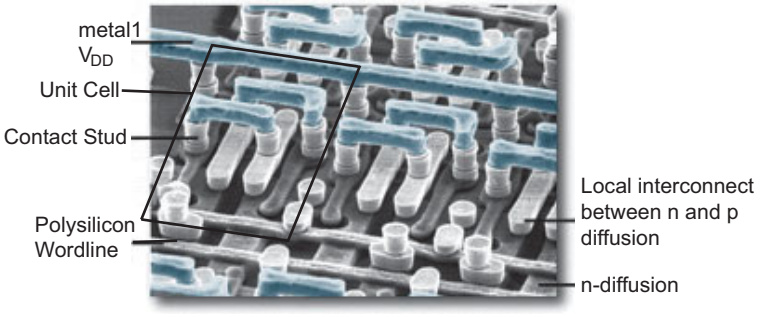


FIGURE 3.12 Partially completed 6-transistor SRAM array using local interconnect (Courtesy of International Business Machines Corporation. Unauthorized use not permitted.)

Contemporary logic processes use copper interconnects and low-k dielectrics to reduce wire resistance and capacitance. These enhancements are discussed in Section 3.4.2.

Figure 3.13 shows a cross-section of an IBM microprocessor showing the 11 layers of metal in a 90 nm process. The bottom level is tungsten local interconnect. The next five layers are on a 1× width and thickness (0.12 μm width and spacing). Metal 6–8 are on a 2× width, spacing, and thickness and metal 9–10 are 4×. These ten layers use copper wires with low-k dielectrics. The top level is aluminum and is primarily used for I/O pads. The local interconnect and metal1 are used in both directions, while the upper layers are used in alternating preferred directions. A pair of vias between metal 9 and 10 are visible. The interfaces between dielectric levels after each step of CMP are also visible.

Figure 3.14 shows a micrograph in which the oxide between metal layers has been stripped away to reveal the complex three-dimensional structure of chip wiring.

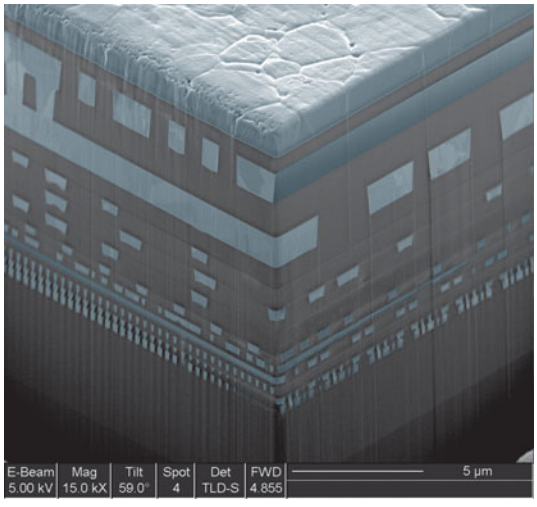


FIGURE 3.13 Cross-section showing 11 levels of metallization (Courtesy of International Business Machines Corporation. Unauthorized use not permitted.)

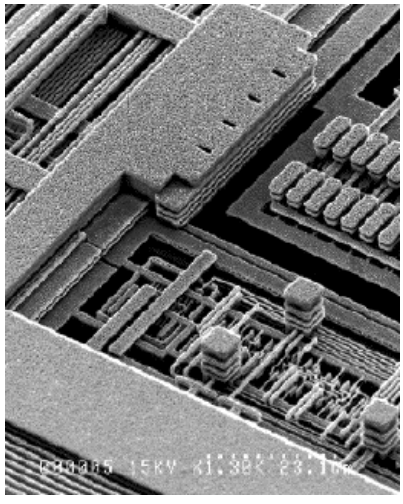


FIGURE 3.14 Micrograph of metallization in six-layer copper process (Courtesy of International Business Machines Corporation. Unauthorized use not permitted.)

3.2.9 Passivation

The final processing step is to add a protective glass layer called *passivation* or *overglass* that prevents the ingress of contaminants. Openings in the passivation layer, called *overglass cuts*, allow connection to I/O pads and test probe points if needed. After passivation, further steps can be performed such as *bumping*, which allows the chip to be directly connected to a circuit board using plated solder bumps in the pad openings.

3.2.10 Metrology

Metrology is the science of measuring. Everything that is built in a semiconductor process has to be measured to give feedback to the manufacturing process. This ranges from simple optical measurements of line widths to advanced techniques to measure thin films and defects such as voids in copper interconnect. A natural requirement exists for *in situ*

real-time measurements so that the manufacturing process can be controlled in a direct feedback manner.

Optical microscopes are used to observe large structures and defects, but are no longer adequate for structures smaller than the wavelength of visible light (~0.5 μm). Scanning electron microscopy (SEM) is used to observe very small features. An SEM raster scans a structure under observation and observes secondary electron emission to produce an image of the surface of the structure. Energy Dispersive Spectroscopy (EDX) bombards a circuit with electrons causing x-ray emission. This can be used for imaging as well. A Transmission Electron Microscope (TEM), which observes the results of passing electrons through a sample (rather than bouncing them off the sample), is sometimes also used to measure structures.

3.3 Layout Design Rules

Layout rules, also referred to as *design rules*, were introduced in Chapter 1 and can be considered a prescription for preparing the photomasks that are used in the fabrication of integrated circuits. The rules are defined in terms of *feature sizes* (widths), *separations*, and *overlaps*. The main objective of the layout rules is to build reliably functional circuits in as small an area as possible. In general, design rules represent a compromise between performance and yield. The more conservative the rules are, the more likely it is that the circuit will function. However, the more aggressive the rules are, the greater the opportunity for improvements in circuit performance and size.

Design rules specify to the designer certain geometric constraints on the layout artwork so that the patterns on the processed wafer will preserve the topology and geometry of the designs. It is important to note that design rules do not represent some hard boundary between correct and incorrect fabrication. Rather, they represent a tolerance that ensures high probability of correct fabrication and subsequent operation. For example, you may find that a layout that violates design rules can still function correctly and vice versa. Nevertheless, any significant or frequent departure (*design rule waiver*) from design rules will seriously prejudice the success of a design.

Chapter 1 described a version of design rules based on the MOSIS CMOS scalable rules. The MOSIS rules are expressed in terms of λ. These rules allow some degree of scaling between processes, as in principle, you only need to reduce the value of λ and the designs will be valid in the next process down in size. Unfortunately, history has shown that processes rarely shrink uniformly. Thus, industry usually uses the actual micron design rules for layouts. At this time, custom layout is usually constrained to a number of often-used standard cells or memories, where the effort expended is amortized over many instances. Only for extremely high-volume chips is the cost savings of a smaller full-custom layout worth the labor cost of that layout.

3.3.1 Design Rule Background

We begin by examining the reasons for the most important design rules.

3.3.1.1 Well Rules The n-well is usually a deeper implant (especially a deep n-well) than the transistor source/drain implants, and therefore, it is necessary to provide sufficient clearance between the n-well edges and the adjacent *n*+ diffusions. The clearance between the well edge and an enclosed diffusion is determined by the transition of the field oxide

across the well boundary. Processes that use STI may permit zero inside clearance. In older LOCOS processes, problems such as the bird's beak effect usually force substantial clearances. Being able to place nMOS and pMOS transistors closer together can significantly reduce the size of SRAM cells.

Because the n-well sheet resistance can be several kΩ per square, it is necessary to ground the well thoroughly by providing a sufficient number of well taps. This will prevent excessive voltage drops due to well currents. Guidelines on well and substrate taps are given in Section 7.3.6. Where wells are connected to different potentials (say in analog circuits), the spacing rules may differ from *equipotential* wells (all wells at the same voltage—the normal case in digital logic).

Mask Summary: The masks encountered for well specification may include n-well, p-well, and deep n-well. These are used to specify where the various wells are to be placed. Often only one well is specified in a twin-well process (i.e., n-well) and by default the p-well is in areas where the n-well isn't (i.e., p-well equals the logical NOT of the n-well).

3.3.1.2 Transistor Rules CMOS transistors are generally defined by at least four physical masks. These are active (also called diffusion, diff, thinox, OD, or RX), n-select (also called n-implant, nimp, or nplus), p-select (also called p-implant, pimp, or pplus) and polysilicon (also called poly, polyg, PO, or PC). The active mask defines all areas where either n- or p-type diffusion is to be placed *or* where the gates of transistors are to be placed. The gates of transistors are defined by the logical AND of the polysilicon mask and the active mask, i.e., where polysilicon crosses diffusion. The select layers define what type of diffusion is required. n-select surrounds active regions where n-type diffusion is required. p-select surrounds areas where p-type diffusion is required. n-diffusion areas inside p-well regions define nMOS transistors (or n-diffusion wires). n-diffusion areas inside n-well regions define n-well contacts. Likewise, p-diffusion areas inside n-wells define pMOS transistors (or p-diffusion wires). p-diffusion areas inside p-wells define substrate contacts (or p-well contacts). Frequently, design systems will define only n-diffusion (ndiff) and p-diffusion (pdiff) to reduce the complexity of the process. The appropriate selects are generated automatically. That is, ndiff will be converted automatically into active with an overlapping rectangle or polygon of n-select.

It is essential for the poly to cross active completely; otherwise the transistor that has been created will be shorted by a diffusion path between source and drain. Hence, poly is required to extend beyond the edges of the active area. This is often termed the *gate extension*. Active must extend beyond the poly gate so that diffused source and drain regions exist to carry charge into and out of the channel. Poly and active regions that should not form a transistor must be kept separated; this results in a spacing rule from active to polysilicon.

Figure 3.15(a) shows the mask construction for the final structures that appear in Figure 3.15(b).

Mask Summary: The basic masks (in addition to well masks) used to define transistors, diffusion interconnect (possibly resistors), and gate interconnect are active, n-select, p-select, and polysilicon. These may be called different names in some processes. Sometimes n-diffusion (ndiff) and p-diffusion (pdiff) masks are used in place of active to alleviate designer confusion.

3.3.1.3 Contact Rules There are several generally available contacts:

- Metal to p-active (p-diffusion)
- Metal to n-active (n-diffusion)

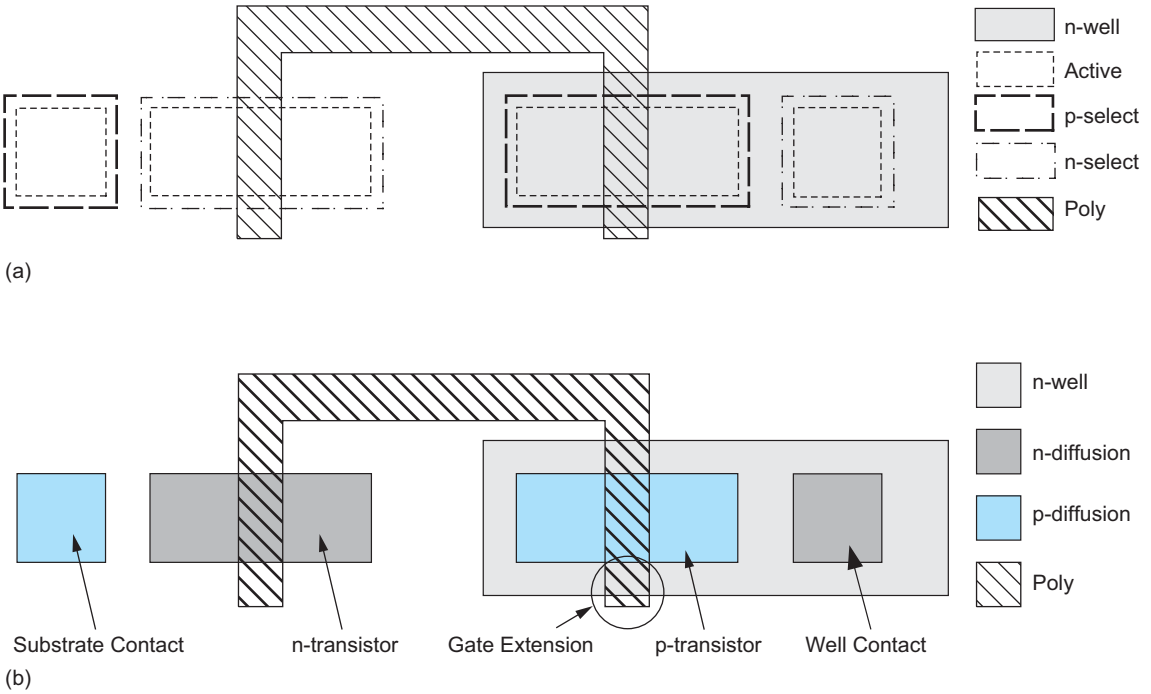


FIGURE 3.15 CMOS n-well process transistor and well/substrate contact construction

- Metal to polysilicon
- Metal to well or substrate

Depending on the process, other contacts such as *buried* polysilicon-active contacts may be allowed for local interconnect.

Because the substrate is divided into well regions, each isolated well must be tied to the appropriate supply voltage; i.e., the n-well must be tied to V_{DD} and the substrate or p-well must be tied to GND with well or substrate contacts. As mentioned in Section 1.5.1, metal makes a poor connection to the lightly doped substrate or well. Hence, a heavily doped active region is placed beneath the contact, as shown at the source of the nMOS transistor in Figure 3.16.

Whenever possible, use more than one contact at each connection. This significantly improves yield in many processes because the connection is still made even if one of the contacts is malformed.

Mask Summary: The only mask involved with contacts to active or poly is the contact mask, commonly called CONT or CA. Contacts are normally of uniform size to allow for consistent etching of very small features.

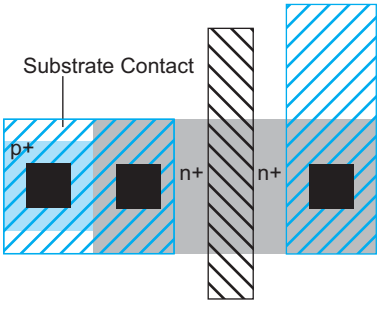


FIGURE 3.16 Substrate contact

3.3.1.4 Metal Rules Metal spacing may vary with the width of the metal line (so called fat-metal rules). That is, above some metal wire width, the minimum spacing may be increased. This is due to etch characteristics of small versus large metal wires. There may also be maximum metal width rules. That is, single metal wires cannot be greater than a certain width. If wider wires are desired, they are constructed by paralleling a number of smaller wires and

adding checkerboard links to tie the wires together. Additionally, there may be spacing rules that are applied to long, closely spaced parallel metal lines.

Older nonplanarized processes required greater width and spacing on upper-level metal wires (e.g., metal3) to prevent breaks or shorts between adjoining wires caused by the vertical topology of the underlying layers. This is no longer a consideration for modern planarized processes. Nevertheless, width and spacing are still greater for thicker metal layers.

Mask Summary: Metal rules may be complicated by varying spacing dependent on width: As the width increases, the spacing increases. Metal overlap over contact might be zero or nonzero. Guidelines will also exist for electromigration, as discussed in Section 7.3.3.1.

3.3.1.5 Via Rules Processes may vary in whether they allow *stacked* vias to be placed over polysilicon and diffusion regions. Some processes allow vias to be placed within these areas, but do not allow the vias to straddle the boundary of polysilicon or diffusion. This results from the sudden vertical topology variations that occur at sublayer boundaries. Modern planarized processes permit stacked vias, which reduces the area required to pass from a lower-level metal to a high-level metal.

Mask Summary: Vias are normally of uniform size within a layer. They may increase in size toward the top of a metal stack. For instance, large vias required on power busses are constructed from an array of uniformly sized vias.

3.3.1.6 Other Rules The passivation or overglass layer is a protective layer of SiO_2 (glass) that covers the final chip. Appropriately sized openings are required at pads and any internal test points.

Some additional rules that might be present in some processes are as follows:

- Extension of polysilicon or metal beyond a contact or via
- Differing gate poly extensions depending on the device length
- Maximum width of a feature
- Minimum area of a feature (small pieces of photoresist can peel off and float away)
- Minimum notch sizes (small notches are rarely beneficial and can interfere with resolution enhancement techniques)

3.3.1.7 Summary Whereas earlier processes tended to be process driven and frequently had long and involved design rules, processes have become increasingly "designer friendly" or, more specifically, computer friendly (most of the mask geometries for designs are algorithmically produced). Companies sometimes create "generic" rules that span a number of different CMOS foundries that they might use. Some processes have design guidelines that feature structures to be avoided to ensure good yields. Traditionally, engineers followed yield-improvement cycles to determine the causes of defective chips and modify the layout to avoid the most common systematic failures. Time to market and product life cycles are now so short that yield improvement is only done for the highest volume parts. It is often better to reimplement a successful product in a new, smaller technology rather than to worry about improving the yield on the older, larger process.

3.3.2 Scribe Line and Other Structures

The *scribe line* surrounds the completed chip where it is cut with a diamond saw. The construction of the scribe line varies from manufacturer to manufacturer. It is designed to

prevent the ingress of contaminants from the side of the chip (as opposed to the top of the chip, which is protected by the overglass).

Several other structures are included on a mask including the *alignment mark*, *critical dimension* structures, *vernier* structures, and *process check* structures [Hess94]. The mask alignment mark is usually placed by the foundry to align one mask to the next. Critical dimension test structures can be measured after processing to check proper etching of narrow polysilicon or metal lines. Vernier structures are used to judge the alignment between layers. A vernier is a set of closely spaced parallel lines on two layers. Misalignment between the two layers can be judged by the alignment of the two verniers. Test structures such as chains of contacts and vias, test transistors, and ring oscillators are used to evaluate contact resistance and transistor parameters. Often these structures can be placed along the scribe line so they do not consume useful wafer area.

3.3.3 MOSIS Scalable CMOS Design Rules

Class project designs often use the λ-based scalable CMOS design rules from MOSIS because they are simple and freely available. MOSIS once offered a wide variety of processes, from 2 μm to 180 nm, compatible with the scalable CMOS rules. Indeed, MOSIS also supports three variants of these rules: SCMOS, SUBM, and DEEP, which are progressively more conservative to support feature sizes down to 180 nm. Chips designed in the conservative DEEP rules could be fabricated on any of the MOSIS processes.

As time has passed, the older processes became obsolete and the newer processes have too many nuances to be compatible with scalable design rules. The MOSIS processes most commonly used today are the ON Semiconductor (formerly AMI) 0.5 μm process and the IBM 130, 90, 65, and 45 nm processes.

The 0.5 μm process is popular for university class projects because MOSIS Educational Program offers generous grants to cover fabrication costs for 1.5 mm $\times$ 1.5 mm "TinyChips." The best design rules for this process are the scalable SUBM rules[1] using $\lambda = 0.3$ μm. Thus, a TinyChip is 5000 $\lambda \times 5000$ λ. Polysilicon is drawn at 2 $\lambda = 0.6$ μm, then biased by MOSIS by –0.1 μm prior to mask generation to give a true 0.5 μm gate length. When simulating circuits, be sure to use the biased channel lengths to model the transistor behavior accurately. In SPICE, the XL parameter is added to the specified transistor length to find the actual length. Thus, a SPICE deck could specify a drawn channel length of L = 0.6 μm for each transistor and include XL = -0.1μm in the model file to indicate a biased length of 0.5 μm. There is a tutorial at `www.cmosvlsi.com` on designing in this process with the Electric CAD tool suite. [Brunvand09] explains how to design in this process with the Cadence and Synopsys tool suites; this flow has a steeper learning curve but better mirrors industry practices.

Credible research chips need more advanced processes to reflect contemporary design challenges. The IBM processes are presently discounted for universities, and MOSIS offers certain research grants as well. The best way to design in these processes is with the Cadence and Synopsys tools using IBM's proprietary micron-based design rules. The design flow is presently poorly documented by MOSIS and ranges from difficult at the 130 nm node to worse at deeper nodes. Unfortunately, this presently limits access to these processes to highly sophisticated research groups.

[1]Technically, MOSIS has two sets of contact rules [MOSIS09]. The standard rules require polysilicon and active to overlap contacts by 1.5 λ. Half-lambda rules reduce productivity because they force the designer off a λ grid. The "alternate contact rules" are preferable because they require overlap by 1 λ, at the expense of more conservative spacing rules; these alternate rules are used in the examples in this text.

Section 1.5.3 introduced the SCMOS design rules. More extensive rules are illustrated and summarized on the inside back cover. Layouts consist of a set of rectangles on various layers such as polysilicon or metal. *Width* is the minimum width of a rectangle on a particular layer. *Spacing* is the minimum spacing between two rectangles on the same or different layers. *Overlap* specifies how much a rectangle must surround another on another layer. Dimensions are all specified in λ except for overglass cuts that do not scale well because they must contact large bond wires or probe tips. Select layers are often generated automatically and thus are not shown in the layout. If the active layer satisfies design rules, the select will too.

Contacts and vias must be exactly 2×2 λ. Larger connections are made from arrays of small vias to prevent current crowding at the periphery. The spacing rules of polysilicon or diffusion to arrays of multiple contacts is slightly larger than that to a single contact.

Section 1.5.5 estimated the pitch of lower-level metal to be 8 λ: 4 λ for the width and 4 λ for spacing. Technically, the minimum width and spacing are 3 λ, but the minimum metal contact size is 2×2 λ plus 1 λ surround on each side, for a width of 4 λ. Thus, the pitch for contacted metal lines can be reduced to 7 λ. Moreover, if the lines are drawn at 3 λ and the contacts are staggered so two adjacent lines never have adjacent contacts, the pitch reduces to 6.5 λ. Nevertheless, using a pitch of 8 λ for planning purposes is good practice and leaves a bit of "wiggle room" to solve difficult layout problems.

3.3.4 Micron Design Rules

Table 3.1 lists a set of micron design rules for a hypothetical 65 nm process representing an amalgamation of several real processes. Rule numbers reference the diagram on the inside back cover. Observe that the rules differ slightly but not immensely from lambda-based rules with $\lambda = 0.035$ μm. A complete set of micron design rules in this generation fills hundreds of pages. Note that upper level metal rules are highly variable depending on the metal thickness; thicker wires require greater widths and spacings and bigger vias.

TABLE 3.1 Micron design rules for 65 nm process

Layer	Rule	Description	65 nm Rule (μm)
Well	1.1	Width	0.5
	1.2	Spacing to well at different potential	0.7
	1.3	Spacing to well at same potential	0.7
Active (diffusion)	2.1	Width	0.10
	2.2	Spacing to active	0.12
	2.3	Source/drain surround by well	0.15
	2.4	Substrate/well contact surround by well	0.15
	2.5	Spacing to active of opposite type	0.25
Poly	3.1	Width	0.065
	3.2	Spacing to poly over field oxide	0.10
	3.2a	Spacing to poly over active	0.10
	3.3	Gate extension beyond active	0.10
	3.4	Active extension beyond poly	0.10
	3.5	Spacing of poly to active	0.07

TABLE 3.1 Micron design rules for 65 nm process (continued)

Layer	Rule	Description	65 nm Rule (μm)
Select	4.1	Spacing from substrate/well contact to gate	0.15
	4.2	Overlap of active	0.12
	4.3	Overlap of substrate/well contact	0.12
	4.4	Spacing to select	0.20
Contact (to poly or active)	5.1, 6.1	Width (exact)	0.08
	5.2b, 6.2b	Overlap by poly or active	0.01
	5.3, 6.3	Spacing to contact	0.10
	5.4	Spacing to gate	0.07
Metal1	7.1	Width	0.09
	7.2	Spacing to well metal1	0.09
	7.3, 8.3	Overlap of contact or via	0.01
	7.4	Spacing to metal for lines wider than 0.5 μm	0.30
Via1–Via6	8.1, 14.1, …	Width (exact)	0.10
	8.2, 14.2, …	Spacing to via on same layer	0.10
Metal2–Metal7	9.1, …	Width	0.10
	9.2, …	Spacing to same layer metal	0.10
	9.3, …	Overlap of via	0.01
	9.4, …	Spacing to metal for lines wider than 0.5 μm	0.30
Via7–8		Width	0.20
		Spacing	0.20
Metal8–9		Width	0.40
		Spacing to same layer metal	0.40
		Overlap of via	0.10
		Spacing to metal for lines wider than 1.0 μm	0.50

3.4 CMOS Process Enhancements

3.4.1 Transistors

3.4.1.1 Multiple Threshold Voltages and Oxide Thicknesses Some processes offer multiple threshold voltages and/or oxide thicknesses. Low-threshold transistors deliver more ON current, but also have greater subthreshold leakage. Providing two or more thresholds permits the designer to use low-V_t devices on critical paths and higher-V_t devices elsewhere to limit leakage power. Multiple masks and implantation steps are used to set the various thresholds. Alternatively, transistors with slightly longer channels can be used; these transistors naturally have higher thresholds because of the short channel effect (see Section 2.4.3.3) [Rohrer05].

Thin gate oxides also permit more ON current. However, they break down when exposed to the high voltages needed in I/O circuits. Oxides thinner than about 15 Å also

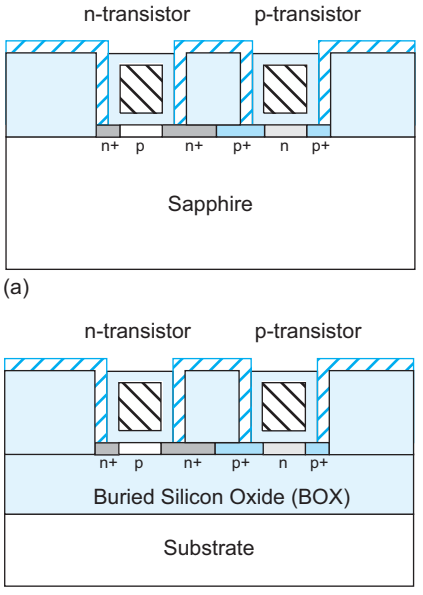


FIGURE 3.17 SOI types

contribute to large gate leakage currents. Many processes offer a second, thicker oxide for the I/O transistors (see Section 13.6). For example, 3.3 V I/O circuits commonly use 0.35 μm channel lengths and 7 nm gate oxides. When gate leakage is a problem and high-k dielectrics are unavailable, an intermediate oxide thickness may also be provided to reduce leakage. Again, multiple masks are used to define the different oxides.

3.4.1.2 Silicon on Insulator A variant of CMOS that has been available for many years is Silicon on Insulator (SOI). As the name suggests, this is a process where the transistors are fabricated on an insulator. SOI stands in contrast to conventional bulk processes in which the transistors are fabricated on a conductive substrate. Two main insulators are used: SiO_2 and sapphire. One major advantage of an insulating substrate is the elimination of the capacitance between the source/drain regions and body, leading to higher-speed devices. Another major advantage is lower subthreshold leakage due to steeper subthreshold slope resulting from a smaller n in EQ (2.44). The drawbacks are time-dependent threshold variations caused by the floating body.

Figure 3.17 shows two common types of SOI. Figure 3.17(a) illustrates a sapphire substrate. In this technology (for example, Peregrine Semiconductor's UltraCMOS), a thin layer of silicon is formed on the sapphire surface. The thin layer of silicon is selectively doped to define different threshold transistors. Gate oxide is grown on top of this and then polysilicon gates are defined. Following this, the nMOS and pMOS transistors are formed by implantation. Figure 3.17(b) shows a silicon-based SOI process. Here, a silicon substrate is used and a *buried oxide* (BOX) is grown on top of the silicon substrate. A thin silicon layer is then grown on top of the buried oxide and this is selectively implanted to form nMOS and pMOS transistor regions. Gate, source, and drain regions are then defined in a similar fashion to a bulk process. Sapphire is optically and RF transparent. As such, it can be of use in optoelectronic areas when merged with III-V based light emitters.

SOI devices and circuits are discussed further in Section 9.5.

3.4.1.3 High-k Gate Dielectrics MOS transistors need high gate capacitance to attract charge to the channel. This leads to very thin SiO_2 gate dielectrics (e.g., 10.5–12 Å, merely four atomic layers, in a 65 nm process). Gate leakage increases unacceptably below these thicknesses, which brings an end to classical scaling [Bai04]. Simple SiO_2 has a dielectric constant of $k = 3.9$. As shown in EQ (2.2), gates could use thicker dielectrics and hence leak less if a material with a higher dielectric constant were available.

A first step in this direction was the introduction of nitrogen to form *oxynitride* gate dielectrics, called SiON, around the 130 nm generation, providing k of about 4.1–4.2. High-k dielectrics entered commercial manufacturing in 2007, first with a hafnium-based material in Intel's 45 nm process [Auth08]. Hafnium oxide (HfO_2) has $k = 20$.

A depletion region forms at the interface of polysilicon and the gate dielectric. This effectively increases t_{ox}, which is undesirable for performance. Moreover, polysilicon gates can be incompatible with high-k dielectrics because of effects such as *threshold voltage pinning* and *phonon scattering*, which make it difficult to obtain low thresholds and reduce the mobility. The Intel 45 nm process returned to metal gates to solve these problems and also to reduce gate resistance, as shown in Figure 3.18 [Mistry07]. Thus, the term MOS is

technically accurate again! nMOS and pMOS transistors use different types of metal with different work functions (energy required to free an electron from a solid) to set the threshold voltages. A second lower-resistance metal layer plays a role similar to a silicide.

One of the challenges with metal gates is that they melt if exposed to the high temperature source/drain formation steps. But if the gate were formed after the source and drain, the self-alignment advantage would be lost. Intel sidesteps this conundrum by first building the transistor with a high-k dielectric and a standard polysilicon gate. After the transistor is complete and the interlayer dielectric is formed, the wafer is polished to expose the polysilicon gates and etched to remove the undesired poly. A thin metal gate is deposited in the trench. Different metals with different workfunctions are required for the nMOS and pMOS transistors. Finally, the trench is filled with a thicker layer of Al for low gate resistance, and the wafer is planarized again.

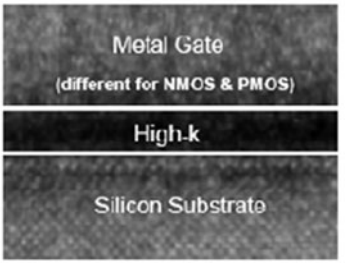


FIGURE 3.18 High-k gate stack TEM (© IEEE 2007.)

3.4.1.4 Higher Mobility Increasing the mobility (μ) of the semiconductor improves drive current and transistor speed. One way to improve the mobility is to introduce mechanical strain in the channel. This is called *strained silicon*.

Figure 3.19 shows strained nMOS and pMOS transistors in the Intel 65 nm process that achieve 40% and 100% higher mobility than unstrained transistors, respectively [Tyagi05, Thompson02, Thompson04]. The nMOS channel is under tensile stress created by an insulating film of silicon nitride (SiN) capping the gate. The pMOS channel is under compressive stress produced by etching a recess into the source and drain, then filling the slot with an epitaxial layer of silicon germanium (SiGe). Germanium is another group IV semiconductor with a larger atomic radius than silicon. When a small fraction of the silicon atoms are replaced by germanium, the lattice retains its shape but undergoes mechanical strain because of the larger atoms. Using separate strain mechanisms for the nMOS and pMOS transistors improves mobility of both electrons and holes. An alternative approach is to implant germanium atoms in the channel, introducing tensile stress that only improves electron mobility. STI also introduces stress that affects mobility, so the diffusion layout can impact performance [Topaloglu07].

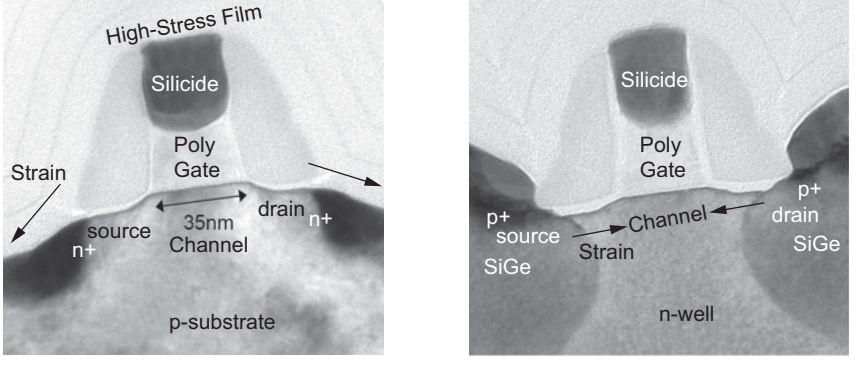


FIGURE 3.19 Strained silicon transistor micrographs: (a) nMOS, (b) pMOS (© IEEE 2005.)

SiGe is also used in high-performance bipolar transistors, especially for radio-frequency (RF) applications. SiGe bipolar devices can be combined with conventional CMOS on the same substrate, which is valuable for low-cost system-on-chip applications that require both digital and RF circuits [Hashimoto02, Harame01a, Harame01b].

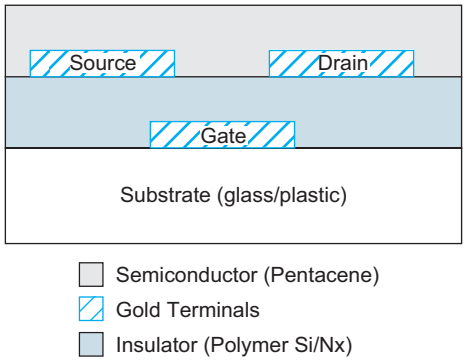


FIGURE 3.20 Plastic transistors

3.4.1.5 Plastic Transistors MOS transistors can be fabricated with organic chemicals. These transistors show promise in active matrix displays, flexible electronic paper, and radio-frequency ID tags because the devices can be manufactured from an inexpensive chemical solution [Huitema03, Myny09]. Figure 3.20 shows the structure of a plastic pMOS transistor. The transistor is built "upside down" with the gold gates and interconnect patterned first on the substrate. Then an organic insulator or silicon nitride is laid down, followed by the gold source and drain connections. Finally, the organic semiconductor (pentacene) is laid down. The mobility of the carriers in the plastic pMOS transistor is about 0.15 $cm^2/V \cdot s$. This is three orders of magnitude lower than that of a comparable silicon device, but is good enough for special applications. Typical lengths and widths are 5 μm and 400 μm, respectively.

3.4.1.6 High-Voltage Transistors High-voltage MOSFETs can also be integrated onto conventional CMOS processes for switching and high-power applications. Gate oxide thickness and channel length have to be larger than usual to prevent breakdown. Specialized process steps are necessary to achieve very high breakdown voltages.

3.4.2 Interconnect

Interconnect has advanced rapidly. While two or three metal layers were once the norm, CMP has enabled inexpensive processes to include seven or more layers. Copper metal and low-k dielectrics are almost universal to reduce the resistance and capacitance of these wires.

3.4.2.1 Copper Damascene Process While aluminum was long the interconnect metal of choice, copper has largely superseded it in nanometer processes. This is primarily due to the higher conductivity of copper compared to aluminum. Some challenges of adopting copper include the following [Merchant01]:

- Copper atoms diffuse into the silicon and dielectrics, destroying transistors.
- The processing required to etch copper wires is tricky.
- Copper oxide forms readily and interferes with good contacts.
- Care has to be taken not to introduce copper into the environment as a pollutant.

Barrier layers have to be used to prevent the copper from entering the silicon surface. A new metallization procedure called the *damascene process* was invented to form this barrier. The process gets its name from the medieval metallurgists of Damascus who crafted fine inlaid swords. In a conventional subtractive aluminum-based metallization step, as we have seen, aluminum is layered on the silicon surface (where vias also have been etched) and then a mask and resist are used to define which areas of metal are to be retained. The unneeded metal is etched away. A dielectric (SiO_2 or other) is then placed over the aluminum conductors and the process can be repeated.

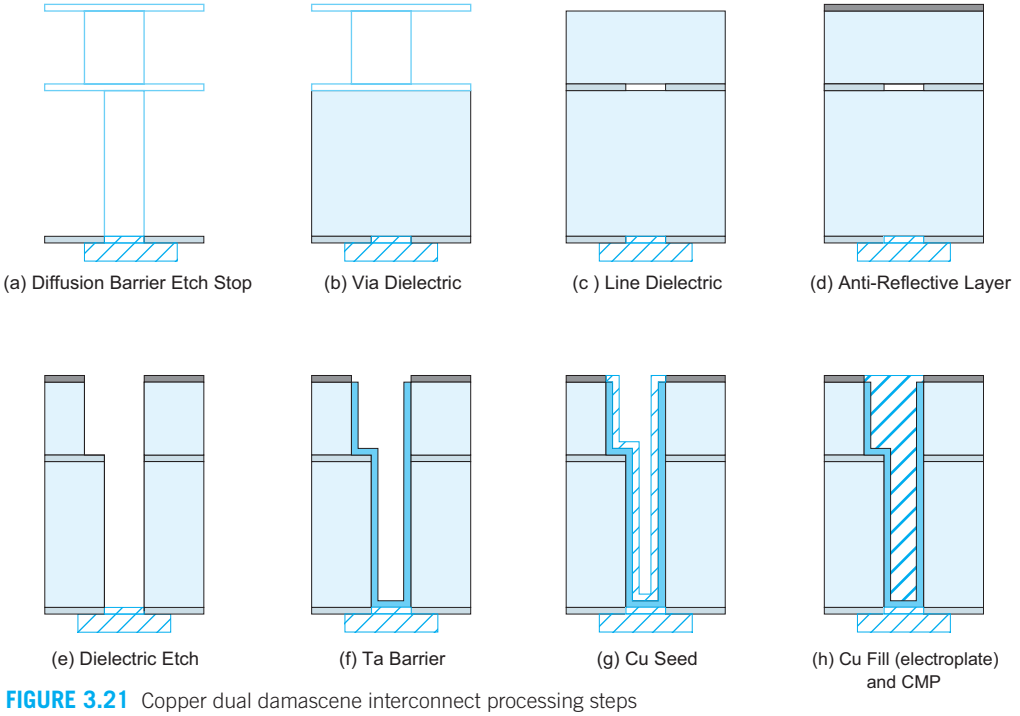


FIGURE 3.21 Copper dual damascene interconnect processing steps

A typical copper damascene process is shown in Figure 3.21, which is an adaptation of a dual damascene process flow from Novellus. Figure 3.21(a) shows a barrier layer over the prior metallization layer. This stops the copper from diffusing into the dielectric and silicon. The via dielectric is then laid down (Figure 3.21(b)). A further barrier layer can then be patterned, and the line dielectric is layered on top of the structure, as shown in Figure 3.21(c). An anti-reflective layer (which helps in the photolithographic process) is added to the top of the sandwich. The two dielectrics are then etched away where the lines and vias are required. A barrier layer such as 10 nm thick Ta or TaN film is then deposited to prevent the copper from diffusing into the dielectrics [Peng02]. As can be seen, a thin layer of the barrier remains at the bottom of the via so the barrier must be conductive. A copper seed layer is then coated over the barrier layer (Figure 3.21(g)). The resulting structure is electroplated full of copper, and finally the structure is ground flat with CMP, as shown in Figure 3.21(h).

3.4.2.2 Low-k Dielectrics SiO_2 has a dielectric constant of $k = 3.9–4.2$. Low-k dielectrics between wires are attractive because they decrease the wire capacitance [Brown03]. This reduces wire delay, noise, and power consumption. Adding fluorine to the silicon dioxide creates *fluorosilicate glass* (FSG or SiOF) with a dielectric constant of 3.6, widely used in 130 nm processes. Adding carbon to the oxide can reduce the dielectric constant to about 2.8–3; such SiCOH (also called *carbon-doped oxide*, CDO) is commonly used at the 90 and 65 nm generation. Alternatively, porous polymer-based dielectrics can deliver even lower dielectric constants. For example, SiLK, from Dow Chemical, has $k = 2.6$ and may scale to $k = 1.6–2.2$ by increasing the porosity. IBM has demonstrated air (or vacuum)

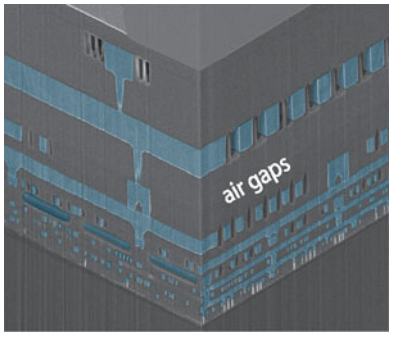


FIGURE 3.22 Micrograph showing air gap insulation between copper wires (Courtesy of International Business Machines Corporation. Unauthorized use not permitted.)

gaps, which have $k = 1$ where the dielectric has been eliminated entirely, as shown in Figure 3.22. Developing low-k dielectrics that can withstand the high temperatures during processing and the forces applied during CMP is a major challenge.

3.4.3 Circuit Elements

While CMOS transistors provide for almost complete digital functionality, the use of CMOS technology as the mixed signal and RF process of choice has driven the addition of special process options to enhance the performance of circuit elements required for these purposes.

3.4.3.1 Capacitors In a conventional CMOS process, a capacitor can be constructed using the gate and source/drain of an MOS transistor, a diffusion area (to ground or V_{DD}), or a parallel metal plate capacitor (using stacked metal layers). The MOS capacitor has good capacitance per area but is relatively nonlinear if operated over large voltage ranges. The diffusion capacitor cannot be used for a floating capacitor (where neither terminal is connected to ground). The metal parallel plate capacitor has low capacitance per area. Normally, the aim in using a floating capacitor is to have the highest ratio of desired capacitance value to stray capacitance (to ground normally). The bottom metal plate contributes stray capacitance to ground.

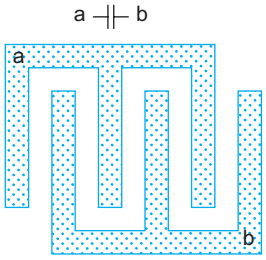


FIGURE 3.23
Fringe capacitor

Analog circuits frequently require capacitors in the range of 1 to 10 pF. The first method for doing this was to add a second polysilicon layer so that a *poly–insulator–poly* (PIP) capacitor could be constructed. A thin oxide was placed between the two polysilicon layers to achieve capacitance of approximately 1 fF/μm^2.

The most common capacitor used in CMOS processes today is a fringe capacitor, which consists of interdigitated fingers of metal, as shown in Figure 3.23. Multiple layers can be stacked to increase the capacitance per area.

3.4.3.2 Resistors In unaugmented processes, resistors can be built from any layer, with the final resistance depending on the resistivity of the layer. Building large resistances in a small area requires layers with high resistivity, particularly polysilicon, diffusion, and n-wells. Diffusion has a large parasitic capacitance to ground, making it unsuitable for high-frequency applications. Polysilicon gates are usually silicided to have low resistivity. The fix for this is to allow for *undoped* high-resistivity polysilicon. This is specified with a *silicide block* mask where high-value poly resistors are required. The resistivity can be tuned to around 300–1000 Ω/square, depending on doping levels. Another material used for precision resistors is nichrome, although this requires a special processing step.

A typical resistor layout is shown in Figure 3.24. This geometry is sometimes called a *meander* structure. A number of *unit* resistors have been used so that a variety of matched resistor values can be constructed. For instance, if 20 kΩ, and 15 kΩ resistors were required, a unit value of 5 kΩ could be used. Then three resistors (as shown) would construct a 15 kΩ resistor. The two resistors at the ends are called *dummy resistors* or fingers. They perform no circuit function, but replicate the proximity effects (such as etch and implant) that the interior resistors see during processing. This helps ensure that all resistors are matched.

The various resistor options have temperature and voltage coefficients. Foundry design manuals normally include these values.

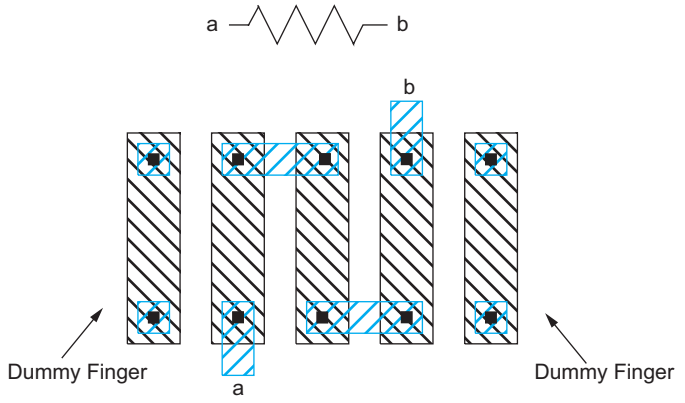


FIGURE 3.24 Resistor layout

3.4.3.3 Inductors The desire to integrate inductors on chips has increased radically with the upsurge in interest in RF circuits. The most common monolithic inductor is the *spiral* inductor, which is a spiral of upper-level metal. A typical inductor is shown in Figure 3.25(a). As the process is planar, an underpass connection has to be made to complete the inductor. A typical equivalent model is shown in Figure 3.25(b). In addition to the required inductance L, there are several parasitic components. R_s is the series resistance of the metal (and contacts) used to form the inductor. C_p is the parallel capacitance to ground due to the area of the metal wires forming the inductor. C_s is the shunt capacitance of the underpass. Finally, R_p is an element that models the loss incurred in the resistive substrate.

Usually, when considering an inductor, the parameters of interest to a designer are its inductance, the Q of the inductor, and the self-resonant frequency. High Qs are sought to create low phase-noise oscillators, narrow filters, and low-loss circuits in general. Q values for typical planar inductors on a bulk process range from 5 to 10.

The number of turns n required to achieve some inductance L if the wire pitch, in turns per meter, is $P = 1/2(W + S)$, is [Lee98]

$$n \approx \sqrt[3]{\frac{LP}{\mu_0}} \tag{3.4}$$

where $\mu_0 = 1.2 \times 10^{-6}$ H/m is the permeability of free space. Figure 3.25 has $n = 1.75$ turns. Higher-quality inductors can also be manufactured using bond wires between I/O pads. The inductance of a wire of length l and radius r is approximately

$$L \approx \frac{\mu_0 l}{2\pi}\left[\ln\frac{2l}{r} - 0.75\right] \tag{3.5}$$

or about 1 nH/mm for standard 1 mil (25 μm) bond wires.

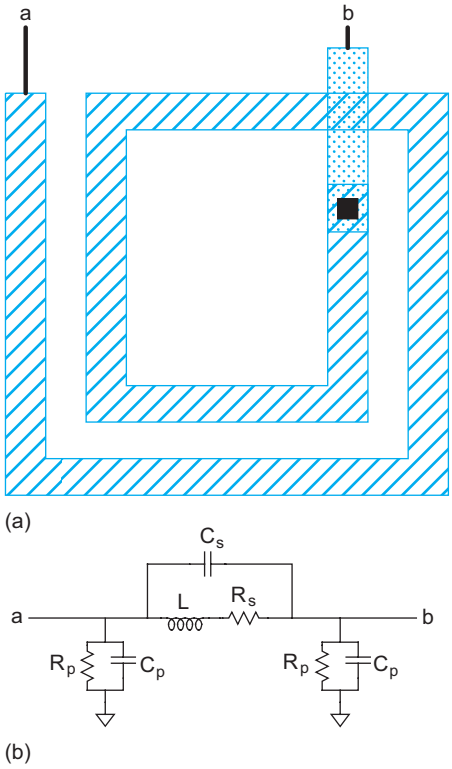


FIGURE 3.25 Typical spiral inductor and equivalent circuit [Rotella02]

Reduction in Q occurs because of the resistive loss in the conductors used to build the inductor (R_s), and the eddy current loss in the resistive silicon substrate (R_p). In an effort to increase Q, designers have resorted to removing the substrate below the inductor using MEMS techniques [Yoon02]. The easiest way to improve the Q of monolithic inductors is to increase the thickness of the top-level metal. The Q can also be improved by using a patterned ground shield in polysilicon under the inductor to decrease substrate losses.

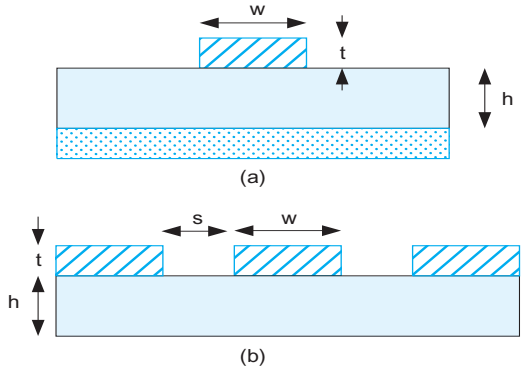


FIGURE 3.26 Microstrip and coplanar waveguide

3.4.3.4 Transmission Lines A *transmission line* can be used on a chip to provide a known impedance wire. Two basic kinds of transmission lines are commonly used: *microstrips* and *coplanar waveguides*.

A microstrip transmission line, as shown in Figure 3.26(a), is composed of a wire of width w and thickness t placed over a ground plane and separated by a dielectric of height h and dielectric constant k. In the chip case, the wire might be the top level of metallization and the ground plane the next metal layer down.

A coplanar waveguide does not require a sublayer ground plane and is shown in Figure 3.26(b). It consists of a wire of width w spaced s on each side from *coplanar* ground wires. The reader is referred to [Wadell91] for detailed design equations.

3.4.3.5 Bipolar Transistors Bipolar transistors were mentioned previously in our discussion of SiGe process options. Both *npn* and *pnp* bipolar transistors can be added to a CMOS process, which is then called a BiCMOS process. These processes tend to be used for specialized analog or high-voltage circuits. In a regular n-well process, a parasitic vertical pnp transistor is present that can be used for circuits such as bandgap voltage references. This transistor is shown in Figure 3.27 with the p-substrate collector, the n-well base, and the p-diffusion emitter. Both process cross-section and layout are shown. This transistor, in conjunction with a parasitic npn, is the cause of latchup (see Section 7.3.6).

3.4.3.6 Embedded DRAM Dynamic RAM (DRAM) uses a single transistor and a capacitor to store a bit of information. It is about five times denser than static RAM (SRAM) conventionally used on CMOS logic chips, so it can reduce the size of a chip containing large amounts of memory. DRAM was conventionally manufactured on specialized processes that produced low-performance logic transistors. DRAM requires specialized structures to build capacitors in a small area. One common structure is a *trench*, which is etched

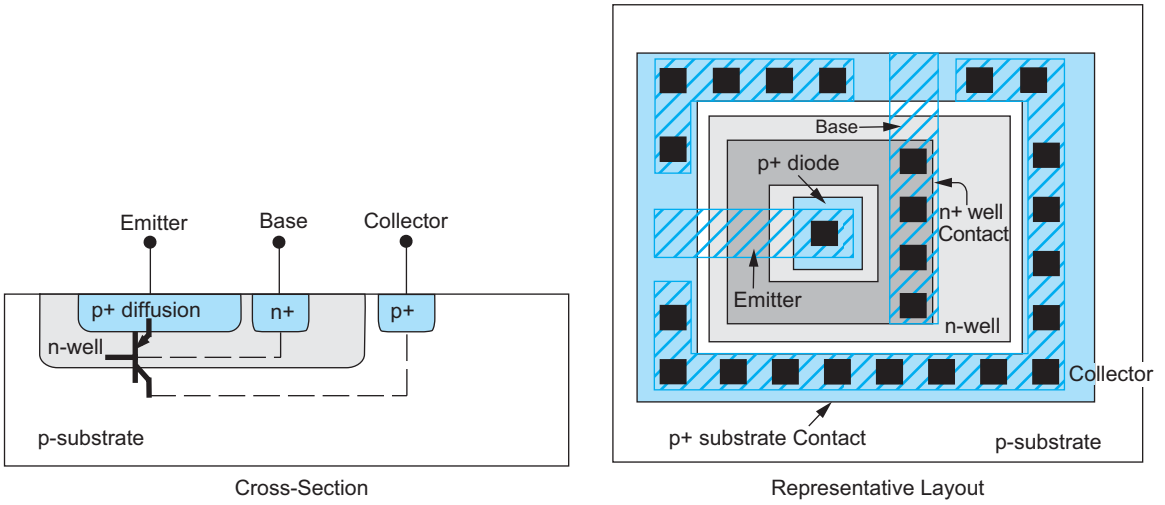


FIGURE 3.27 Vertical pnp bipolar transistor

down into the substrate. Some recent processes have introduced compact capacitor structures for building *embedded DRAM* alongside high-performance logic. Section 12.3 discusses DRAM in more depth.

3.4.3.7 Non-Volatile Memory *Non-volatile* memory (NVM) retains its state when the power is removed from the circuit. The simplest NVM is a *mask-programmed* ROM cell (see Section 12.4). This type of NVM is not reprogrammable or programmable after the device is manufactured. A *one-time programmable* (OTP) memory can be implemented using a fuse constructed of a thin piece of metal through which is passed a current that vaporizes the metal by exceeding the current density in the wire. The first reprogrammable memories used a stacked polysilicon gate structure and were programmed by applying a high voltage to the device in a manner that caused Fowler-Nordheim tunneling to store a charge on a floating gate. The whole memory could be erased by exposing it to UV light that knocked the charge off the gate. These memories evolved to *electrically erasable* memories, which are today represented by *Flash* memory.

A typical Flash memory transistor is shown in Figure 3.28 [She02]. The source and drain structures can vary considerably to allow for high-voltage operation, but the dual-gate structure is fairly common. The gate structure is a stacked configuration commencing with a thin tunnel oxide or nitride. A floating polysilicon gate sits on top of this oxide and a conventional gate oxide is placed on top of the floating gate. Finally, a polysilicon control gate is placed on top of the gate oxide. The operation of the cell is also shown in Figure

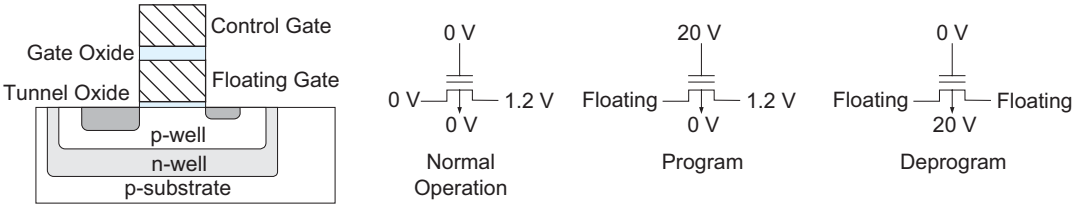


FIGURE 3.28 Flash memory construction and operation

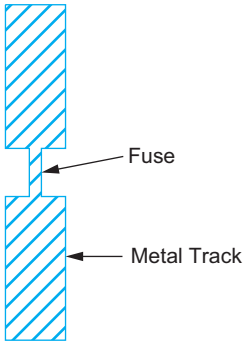


FIGURE 3.29
A typical metal fuse

3.28. In normal operation, the floating gate determines whether or not the transistor is conducting. To program the cell, the source is left floating and the control gate is raised to approximately 20 V (using an on-chip voltage multiplier). This causes electrons to tunnel into the floating gate, and thus program it. To deprogram a cell, the drain and source are left floating and the substrate (or well) is connected to 20 V. The electrons stored on the floating gate tunnel away, leaving the gate in an unprogrammed state.

3.4.3.8 Fuses and Antifuses During manufacturing, fuses can be blown with a high current or zapped by a laser. In the latter case, an area is normally left in the passivation oxide to allow the laser direct access to the metal link that is to be cut. Figure 3.29 shows the layout of a metal fuse.

Laser-blown fuses are large and the blow process can damage adjacent devices. *Electronic fuses* are structures whose characteristics can be nondestructively altered by applying a high current. For example, IBM eFUSEs are narrow polysilicon wires silicided with cobalt. The resistance is initially about 200 Ω. If a programming current of 10–15 mA is applied for 200 μs, the cobalt will migrate to the anode, as shown in Figure 3.30. This raises the resistance by an order of magnitude. Simple sense circuits are used to detect the state of the eFUSE. IBM uses fuses for chip serial numbers, thermal sensor calibration, and to reconfigure defective components [Rohrer05, Rizzolo07].

An *antifuse* is a similar device that initially has a high resistivity but can become low resistance when a programming voltage is applied. This device requires special processing and is used in programmable logic devices (see Section 12.7).

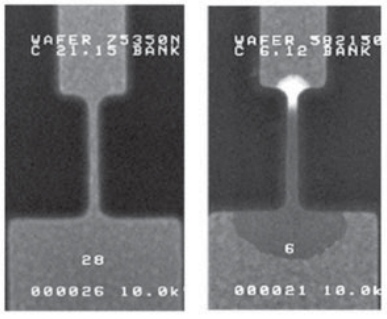


FIGURE 3.30 eFUSE (© IEEE 2005.)

3.4.3.9 Microelectromechanical Systems (MEMS) Semiconductor processes and especially CMOS processes have been used to construct tiny mechanical systems monolithically. A typical device is the well-known air-bag sensor, which is a small accelerometer consisting of an air bridge capacitor that can detect sudden changes in acceleration when co-integrated with some conditioning electronics. MEMS micromirrors on torsional hinges are used in inexpensive, high-resolution digital light projectors. Structures such as cantilevers, mechanical resonators, and even micromotors have been built. A full discussion of MEMS is beyond the scope of this book, but further material can be found in texts such as [Maluf04].

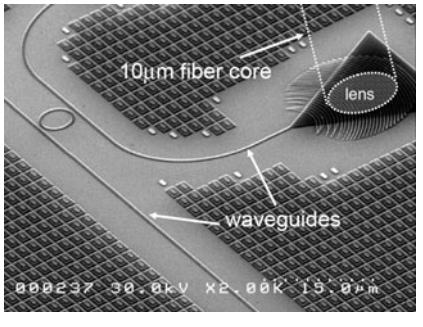


FIGURE 3.31 Optical waveguide and holographic lens integrated with a 130 nm CMOS process (© IEEE 2006.)

3.4.3.10 Integrated Photonics Although silicon is opaque at visible wavelengths, it is transparent in the infrared range used in optical fibers. Semiconductor photonic components are rapidly evolving. Components compatible with a conventional CMOS process include waveguides, modulators, and photodetectors [Salib04, Young10]. A key missing component is an optical source, such as a laser. However, just as V_{DD} is generated off-chip from a DC power supply, light can be generated off-chip and brought onto the chip through an optical fiber. Figure 3.31 shows a holographic lens used to couple an optical fiber to an on-chip waveguide [Huang06]. Integrated photonics shows particular promise for optical transceivers to replace copper wires in high-speed networks.

3.4.3.11 Three-Dimensional Integrated Circuits 3D ICs contain multiple layers of devices. Stacking transistors in layers can reduce wire lengths, improving speed and power. It also can permit heterogeneous technologies to be combined in one package; for example, logic, memory, and analog/RF chips can be stacked into one package.

IBM has described a process in which 200 mm wafers are ground down to a remarkable 20 μm thickness after fabrication [Topol06]. They are aligned to 1 μm tolerance, and then one is bonded on top of another using oxide-fusion or copper bonding. Tall skinny through-silicon vias (TSVs) between the wafers are etched and metallized; the aspect ratio of the vias and the thickness of the wafers sets the density of contacts between wafers. Densities of 10^4 TSVs/mm^2 or more can presently be achieved. Some of the challenges in 3D integration include wafer bowing, testing layers before they are bonded, and managing cooling and power delivery. Figure 3.32 shows two wafers bonded together. The bottom wafer has four levels of metal and the top wafer has two levels. The 8-μm wide landing pad on the top metal layer of the bottom wafer provides tolerances for misalignment with the 3D vias protruding from beneath the top wafer.

3D ICs are starting to move from research into production [Emma08]. An initial application is to stack multiple memory chips to provide a higher capacity in a standard form factor.

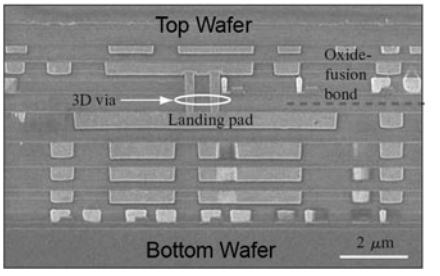


FIGURE 3.32 Scanning electron micrograph of 3-dimensional integration of two wafers (Reprinted from [Koester08]. Courtesy of International Business Machines Corporation. Unauthorized use not permitted.)

3.4.4 Beyond Conventional CMOS

A major problem with scaling bulk transistors is the subthreshold leakage from drain to source caused by the inability of the gate to turn off the channel completely. This can be improved by a gate structure where the gate is placed on two, three, or four sides of the channel to gain better control over the charge in the channel. A promising structure solves the problem by forming a vertical channel and constructing the gate in a pincer-like arrangement around three sides. These devices have been given the generic name "*finfets*" because the source/drain regions form fins on the silicon surface [Hisamoto98]. Figure 3.33(a) shows a 3D view of a finfet, while Figure 3.33(b) shows the cross-section and Figure 3.33(c) shows the top view. The gate wraps around three sides of the vertical

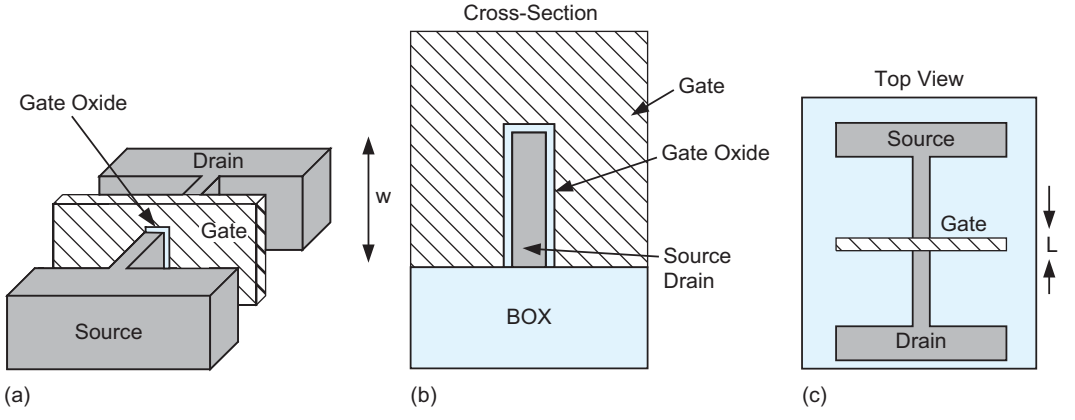


FIGURE 3.33 Finfet structure

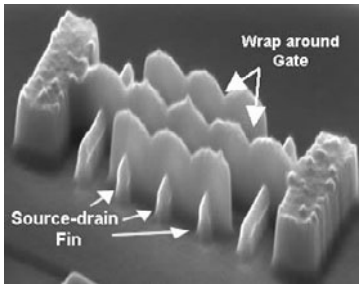


FIGURE 3.34 Trigate transistor (Reprinted with permission of Intel Corporation.)

source/drain fins. The width of the device is defined by the height of the fin, so wide devices are constructed by paralleling fins. Figure 3.34 shows a micrograph of a prototype finfet that Intel calls a *trigate transistor* [Kavalleros06].

Compounds from groups III and V of the periodic table, such as GaAs, offer electron mobilities up to 30 times higher than silicon. Such III-V materials have been research topics for decades. GaAs was once used for very high frequency applications, but has largely been replaced by advanced CMOS processes. However, III-V materials might be integrated into CMOS some day in the future.

Nanotechnology is presently a hot research area seeking alternative structures to replace CMOS when conventional scaling finally runs out of steam. Little obvious progress in radical new device structures has been made since the previous edition of the book, but conventional sub-100 nm CMOS transistors are now being called nanotechnology! Alternative technologies have a large hurdle to overcome competing with the hundreds of billions of dollars that have been invested in advancing CMOS over four decades.

Carbon nanotubes are one nanotechnology that have been used to demonstrate transistor behavior and build inverters [Liu01]. Nanotubes are cylinders with a diameter of a few nanometers. They are of interest because the nanotube is smaller than the predicted endpoint for CMOS gate lengths, and because the nanotubes offer high mobility. A theoretical nanotube transistor is shown in Figure 3.35 [Wong03]. Presently, the speeds are quite slow and the manufacturing techniques are limited, but they may be of interest in the future [Raychowdhury07, Patil09].

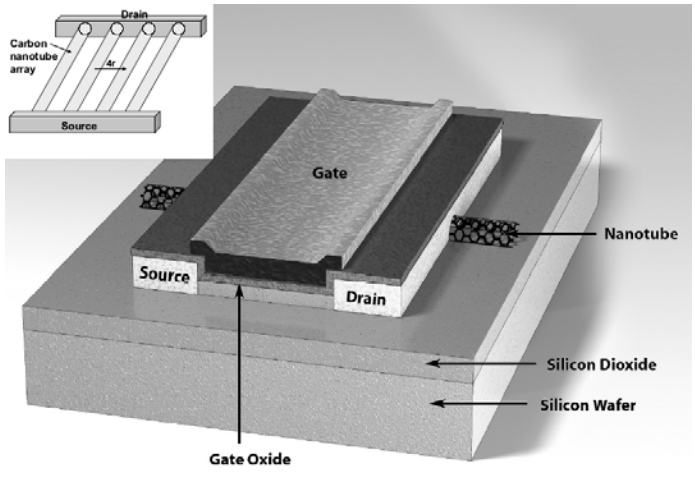


FIGURE 3.35 Carbon nanotube transistor (© IEEE 2003.)

3.5 Technology-Related CAD Issues

The mask database is the interface between the semiconductor manufacturer and the chip designer. Two basic checks have to be completed to ensure that this description can be turned into a working chip. First, the specified geometric design rules must be obeyed. Second, the interrelationship of the masks must, upon passing through the manufacturing

process, produce the correct interconnected set of circuit elements. To check these two requirements, two basic CAD tools are required: a *Design Rule Check* (DRC) program and a mask circuit *extraction* program. The most common approach to implementing these tools is a set of subprograms that perform general geometry operations. A particular set of DRC rules or extraction rules for a given CMOS process (or any semiconductor process) defines the operations that must be performed on each mask and the inter-mask checks that must be completed. Accompanied by a written description, these *run sets* are usually the defining specification for a process.

In this section, we will examine a hypothetical DRC and extraction system to illustrate the nature of these run sets.

3.5.1 Design Rule Checking (DRC)

Although we can design the physical layout in a certain set of mask layers, the actual masks used in fabrication can be derived from the original specification. Similarly, when we want a program to determine what we have designed by examining the interrelationship of the various mask layers, it may be necessary to determine various logical combinations between masks.

To examine these concepts, let us posit the existence of the following functions (loosely based on the Cadence DRACULA DRC program), which we will apply to a geometric database (i.e., rectangles, polygons, and paths):

`AND` `layer1 layer2 -> layer3`
ANDs layer1 and layer2 together to produce layer3
(i.e., the intersection of the two input mask descriptions)

`OR` `layer1 layer2 -> layer3`
ORs layer1 and layer2 together to produce layer3
(i.e., the union of the two input mask descriptions)

`NOT` `layer1 layer2 -> layer3`
Subtracts layer2 from layer1 to produce layer3
(i.e., the difference of the two input mask descriptions)

`WIDTH` `layer > dimension -> layer3`
Checks that all geometry on layer is larger than dimension
Any geometry that is not is placed in layer3

`SPACE` `layer > dimension -> layer3`
Checks that all geometry on layer is spaced further than dimension
Any geometry that is not is placed in layer3

The following layers will be assumed as input:

```
nwell
active
p-select
n-select
poly
poly-contact
active-contact
metal
```

Typically, useful sublayers are generated initially. First, the four kinds of active area are isolated. The rule set to accomplish this is as follows:

```
NOT  all nwell -> substrate
AND  nwell active -> nwell-active
NOT  active nwell -> pwell-active
AND  nwell-active p-select -> pdiff
AND  nwell-active n-select -> vddn
AND  pwell-active n-select -> ndiff
AND  pwell-active p-select -> gndp
```

In the above specification, a number of new layers have been designated. For instance, the first rule states that wherever nwell is absent, a layer called substrate exists. The second rule states that all active areas within the nwell are nwell-active. A combination of nwell-active and p-select or n-select yields pdiff (p diffusion) or vddn (well tap).

To find the transistors, the following rule set is used:

```
AND poly ndiff -> ngates
AND poly pdiff -> pgates
```

The first rule states that the combination of poly and ndiff yields the ngates region—all of the n-transistor gates.

Typical design rule checks (DRC) might include the following:

```
WIDTH metal  < 0.13 -> metal-width-error
SPACE metal  < 0.13 -> metal-space-error
```

For instance, the first rule determines if any metal is narrower than 0.13 μm and places the errors in the metal-width-error layer. This layer might be interactively displayed to highlight the errors.

3.5.2 Circuit Extraction

Now imagine that we want to determine the electrical connectivity of a mask database. The following commands are required:

`CONNECT` `layer1 layer2`
Electrically connect layer1 and layer2.

`MOS` `name drain-layer gate-layer source-layer substrate-layer`
Define an MOS transistor in terms of the component terminal layers. (This is, admittedly, a little bit of magic.)

The connections between layers can be specified as follows:

```
CONNECT active-contact pdiff
CONNECT active-contact ndiff
CONNECT active-contact vddn
CONNECT active-contact gndp
CONNECT active-contact metal
CONNECT gndp substrate
CONNECT vddn nwell
CONNECT poly-contact poly
CONNECT poly-contact metal
```

The connections between the diffusions and metal are specified by the first seven statements. The last two statements specify how metal is connected to poly.

Finally, the active devices are specified in terms of the layers that we have derived:

```
MOS nmos ndiff ngates ndiff substrate
MOS pmos pdiff pgates pdiff nwell
```

An output statement might then be used to output the extracted transistors in some netlist format (i.e., SPICE format). The extracted netlist is often used to compare the layout against the intended schematic.

It is important to realize that the above run set is manually generated. The data you extract from such a program is only as good as the input. For instance, if parasitic routing capacitances are required, then each layer interaction must be coded. If parasitic resistance is important in determining circuit performance, it also must be specifically included in the extraction run set.

3.6 Manufacturing Issues

As processes have evolved, various rules and guidelines have emerged that reflect the complexity of the processing. These rules are often called *Design for Manufacturability* (DFM).

3.6.1 Antenna Rules

When a metal wire contacted to a transistor gate is plasma-etched, it can charge up to a voltage sufficient to zap the thin gate oxides. This is called *plasma-induced gate-oxide damage*, or simply the *antenna effect*. It can increase the gate leakage, change the threshold voltage, and reduce the life expectancy of a transistor. Longer wires accumulate more charge and are more likely to damage the gates.

During the high-temperature plasma etch process, the diodes formed by source and drain diffusions can conduct significant amounts of current. These diodes bleed off charge from wires before gate oxide is damaged.

Antenna rules specify the maximum area of metal that can be connected to a gate without a source or drain to act as a discharge element. Larger gates can withstand more charge buildup. The design rules normally define the maximum ratio of metal area to gate area such that charge on the metal will not damage the gate. The ratios can vary from 100:1 to 5000:1 depending on the thickness of the gate oxide (and hence breakdown voltage) of the transistor in question. Higher ratios apply to thicker gate oxide transistors (i.e., 3.3 V I/O transistors).

Figure 3.36 shows an antenna rule violation and two ways to fix it. In Figure 3.36(a), a long metal1 line is connected to a transistor gate. It has no connection to diffusion until metal2 is formed, so the gate may be damaged during the metal1 plasma etch. In Figure 3.36(b), the metal1 line is interrupted with a jumper to metal2. This reduces the amount of charge that could zap the gate during the metal1 etch and solves the problem. In Figure 3.36(c), an antenna diode is added, providing a discharge path during the etch. The diode is reverse-biased during normal operation and thus does not disturb circuit function (except for the area and capacitance that it contributes). Note that the problem could also have been solved by making the gate wider.

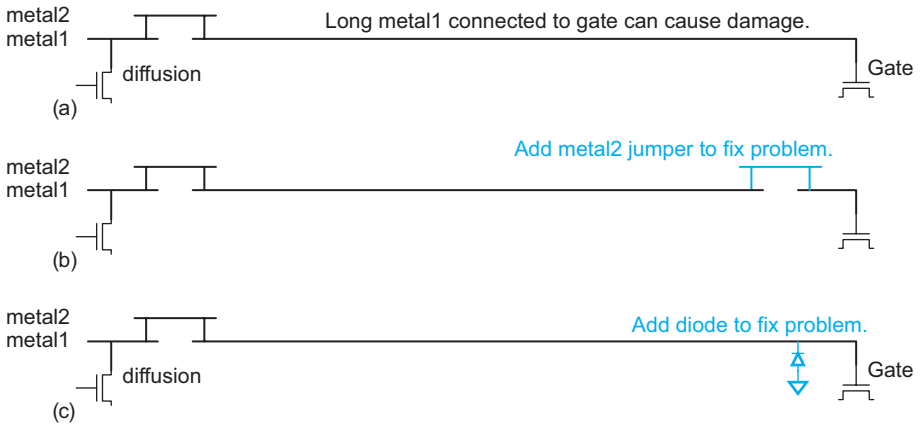


FIGURE 3.36 Antenna rule violation and fixes

For circuits requiring good matching, such as analog and memory cells, transistor gates should connect directly to diffusion with a short segment of metal1 to avoid gate damage that could introduce mismatches.

3.6.2 Layer Density Rules

Another set of rules that pertain to advanced processes are layer density rules, which specify a minimum and maximum density of a particular layer within a specified area. Etch rates have some sensitivity to the amount of material that must be removed. For example, if polysilicon density were too high or too low, transistor gates might end up over- or under-etched, resulting in channel-length variations. Similarly, the CMP process may cause dishing (excessive removal) of copper when the density is not uniform.

To prevent these issues, a metal layer might be required to have 30% minimum and 70% maximum density within a 100 μm by 100 μm area. For digital circuits, these density levels are normally reached with routine routing unless empty spaces exist. Analog and RF circuits, on the other hand, are almost by definition sparse. Thus, diffusion, polysilicon, and metal layers may have to be added manually or by a fill program after design has been completed. The fill can be grounded or left floating. Floating fill contributes lower total capacitance but more coupling capacitance to nearby wires. Grounded fill requires routing the ground net to the fill structures. Clever fill patterns such as staggered rectangles, plus-sign patterns, or diamonds result in lower and more predictable capacitance than do simple geometrical grids [Kahng08]. Designers must be aware of the fill so that it does not introduce unexpected parasitic capacitance to nearby wires.

3.6.3 Resolution Enhancement Rules

Some resolution enhancement techniques impose further design rules. For example, polysilicon typically uses the narrowest lines and thus needs the most enhancement. This can be simplest if polysilicon gates are only drawn in a single orientation (horizontal or vertical). Using a single orientation also reduces systematic process variability. Avoid small jogs and notches (those less than the minimum layer width), because such notches can interfere with proper OPC analysis.

The design community is presently debating a move toward *restrictive design rules* to facilitate RET and reduce manufacturing variability by limiting designers to a smaller set of uniform layout features. These rules might come at the expense of greater area. For example, Intel introduced restrictive design rules for polysilicon in the 45 nm process to control variation and facilitate 193 nm double-patterning lithography [Webb08]. Under these rules, polysilicon is limited to one pitch and direction in layout. This also simplified contact and metal1 rules: the contact pitch is the same as the gate pitch, and metal1 parallel to the gates also has the same pitch. Wide poly pads for contacts and orthogonal polysilicon routing were eliminated by introducing a *trench contact* suitable for local interconnect. Intel found that the restrictive rules did not impact standard cell density and that excellent yield is achieved.

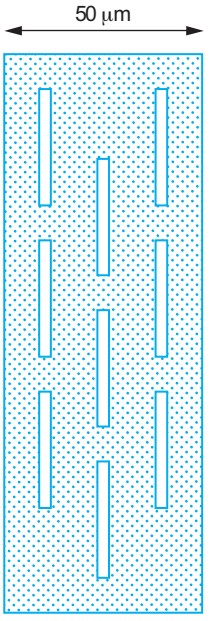


FIGURE 3.37 Slots in wide metal power bus

3.6.4 Metal Slotting Rules

Some processes have special rules requiring that wide (e.g. > 10–40 μm) metal wires have slots. Slots are long slits, on the order of 3 μm wide, in the wire running parallel to the direction of current flow, as shown in Figure 3.37. They provide stress relief, help keep the wire in place, and reduce the risk of electromigration failure (see Section 7.3.3.1). Design rules vary widely between manufacturers.

3.6.5 Yield Enhancement Guidelines

To improve yield, some processes recommend increasing certain widths and spacings where they do not impact area or performance. For example, increasing the polysilicon gate extension slightly reduces the risk of transistor failures from poly/diffusion mask misalignment. Increasing space between metal lines where possible reduces the risk of shorts and also reduces wire capacitance. Other good practices to improve yield include the following:

- Space out wires to reduce risk of short circuits and reduce capacitance.
- Use non-minimum-width wires to reduce risk of open circuits and to reduce resistance.
- Use at least two vias for every connection to avoid open circuits if one via is malformed, and to reduce electromigration wearout.
- Surround contacts and vias by landing pads with more than the minimum overlap to reduce resistance variation and open circuits caused by misaligned contacts.
- Use wider-than-minimum transistors; minimum-width transistors are subject to greater variability and tend not to perform as well.
- Avoid non-rectangular shapes such as 45-degree angles and circles. For specialized circuits such as RAMs that strongly benefit from 45-degree angles, verify masks after optical proximity correction analysis.
- Place dummy transistors or cells at the edge of arrays and sensitive circuits to improve uniformity and matching.
- If it looks nice, it will work better.

3.7 Pitfalls and Fallacies

Targeting a bleeding-edge process

There is a fine balance when you are deciding whether or not to move to a new process for a new design. On the one hand, you are tempted by increased density and speed. On the other hand, support for the new process can initially be expensive (becoming familiar with process rules, CAD tool scripts, porting analog and RF designs, locating logic libraries, etc.). In addition, CMOS foundries frequently tune their processes in the first few months of production, and often yield improvement steps can reflect back to design rule changes that impact designs late in their tapeout schedule. For this reason, it is frequently prudent not to jump immediately into a new process when it becomes available. On the other hand, if you are limited in speed or some other attribute that is solved by the new process, then you don't have much choice but to bite the bullet.

Using lambda design rules on commercial designs

Lambda rules have been used in this text for ease of explanation and consistency. They are usable for class designs. However, they are not very useful for production designs for deep submicron processes. Of particular concern are the metal width and spacing rules, which are too conservative for most production processes.

Failing to account for the parasitic effects of metal fill

With area density rules, particularly in metal, most design flows include an automatic fill step to achieve the correct metal density. Particularly in analog and RF circuits, it is important to either exclude the automatic fill operation from that area or check circuit performance after the fill by completing a full parasitic extract and rerunning the verification simulation scripts.

Failing to include process calibration test structures

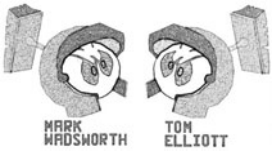
In the discussion on scribe line structures, it was mentioned that test structures are frequently inserted here by the silicon manufacturer. Documentation is often unavailable, so it is prudent for designers (particularly in academic designs, which receive less support from a foundry) to include their own test structures such as transistors or ring oscillators. This allows designers to calibrate the silicon against simulation models.

Waiving design rules

Sometimes it is tempting to ignore a design rule when you are certain it does not apply. For example, consider two wires separated by only 2 λ. This violates a design rule because the wires might short together during manufacturing. If the wires are actually connected elsewhere, one might ignore the rule because further shorting is harmless. However, it is possible that the "antifeature" between the wires would produce a narrow strip of photoresist that could break off and float around during manufacturing, damaging some other structure. Moreover, even if the rule violation is safe, keeping track of all the legitimate exceptions is too much work, especially on a large design. It is better to simply fix the design rule error.

Placing cute logos on a chip

Designers have a tradition of hiding their initials on the chip or embedding cute logos in an unused corner of the die. Some automatic wafer inspection tools find that the logos look more like a spec of dust than a legitimate chip structure and mark all of the chips as defective! Some companies now ban the inclusion of layout that is not essential to the operation of the device. Others require placing the logo in the corner of the chip and covering it with a special pseudolayer called LOGO to tell RET and wafer inspection tools to ignore the logo.



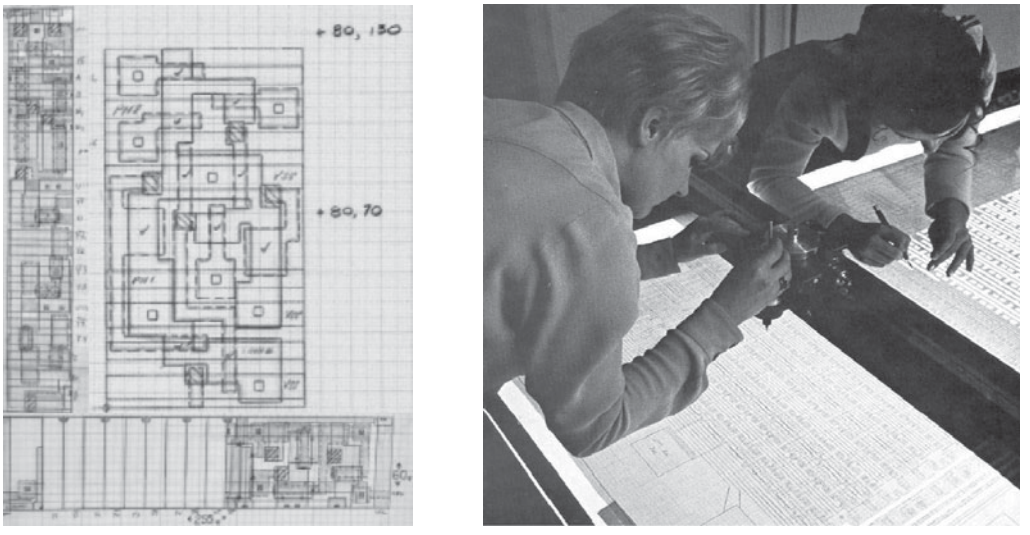
Logos on the image sensor for the Spirit Mars rover (Reprinted from Molecular Expressions Silicon Zoo, `micro.magnet.fsu.edu/creatures`, with permission of Michael Davidson.)

3.8 Historical Perspective

In the first days of integrated circuits, layout editors and design rule checkers were humans with knives and magnifying lenses. [Volk01] tells a captivating story of design at Intel in the early 1970s. Mask designers drew layout with sharp colored pencils on very large sheets of Mylar graph paper, as shown in Figure 3.38(a). Engineers and technicians then scrutinized the drawings to see if all of the design rules were satisfied and if the connections matched the schematic. Most chips at the time were probably manufactured with minor design rule errors, but correct wiring was essential. For example, two engineers each checked all 20,000 transistors on the 8086 in 1977 by hand. Both found 19 of the same 20 errors, giving confidence that the design was correct.

Technicians working at a light table then cut each level of layout onto sheets of *rubylith* to make the masks, as shown in Figure 3.38(b). Rubylith is a two-layered material with a base of heavy transparent Mylar and a thin film of red cellophane-like material. The red film was then peeled away where transistors or wires should be formed. The designer and technician spent days inspecting the rubylith for peeling errors and unintended cuts. The sheets had to be handled with great care to avoid rubbing off pieces. Corrections were performed with a surgical scalpel and metal ruler to add new wires, or with red tape to remove objects. The final result was checked with a 7 times magnifying glass. Finally, the rubylith sheets were sent to a mask vendor to be optically reduced to form the masks. Despite all this care, the initial version of Intel's first product, the 3101 64-bit RAM, was actually a 63-bit RAM because of an error peeling the rubylith. Designers today still gripe at their tools, but the industry has come a long way.

Advances in semiconductor devices are usually presented at the International Electron Devices Meeting (IEDM). Table 3.2 summarizes key characteristics from Intel and IBM.



(a) (b)

FIGURE 3.38 Hand-drawn layout: (a) standard cell, (b) cutting patterns onto rubylith (Reprinted from [Volk01] with permission of Intel Corporation.)

Process development has become so expensive that IBM has formed the Common Platform alliance with partners including Chartered Semiconductor, Samsung, Infineon, and STMicro, to share the R&D costs. IBM offers both SOI and bulk processes; Table 3.2 focuses on their SOI devices that have better I_{dsat} / I_{off} ratios. All of the processes in this table are considered *high-performance processes* that focus on a high I_{dsat}. Many manufacturers also offer *low-power processes* with higher threshold voltages and thicker oxides to reduce leakage, especially in battery-powered communications devices.

TABLE 3.2 CMOS process characteristics

Manufacturer		Intel							IBM			
Feature Size f	nm	250	180	130	90	65	45	32	130	90	65	45
Reference		[Bohr96]	[Yang98]	[Tyagi00]	[Thompson02]	[Bai04]	[Mistry07]	[Natarajan08]	[Sleight01]	[Khare02]	[Lee05]	[Narasimha06]
V_{DD}	V	1.8	1.5	1.3	1.2	1.2	1	1	1.2	1	1	1
L_{gate}	nm	180	140	70	50	35	35	30	60	45	40	35
t_{ox}	nm	4.1	3	1.5	1.2	1.2	1	0.9	2.3	1.85	1.05	1.15
$I_{dsat\text{-}n}$	μA/μm	700	940	1170	1449	1460	1360	1550	915	1000	1137	1140
$I_{dsat\text{-}p}$	μA/μm	320	420	600	725	880	1070	1210	520	480	700	800
I_{off}	nA/μm	1	3	100	400	100	100	100	100	200	200	200
Strain		no	no	no	yes	yes	yes	yes	no	no	yes	yes
High-k Gates		no	no	no	no	no	yes	yes	no	no	no	no
Gate Pitch	nm	640	480	336	260	220	160	112.5	325	245		190
Metal1 Pitch	nm	640	500	350	220	210	150	112.5	350	245		140
Metal Layers		5	6	6	7	8	9	9	8	10	10	10
Material		Al	Al	Cu	Cu	Cu	Cu	Cu	Cu	Cu	Cu	Cu
Low-k Dielectric		none	FSG	FSG	CDO	CDO	CDO	CDO	SiLK	SiLK	CDO	Porous
k		3.9	3.55	3.6	2.9	2.9					2.75	2.4
SRAM Cell Size	μm^2	10.26	5.59	2.09	1	0.57	0.346	0.171	1.8	0.99	0.65	0.37

The transistor characteristics are listed for low-V_t transistors. Since the 130 nm generation, nearly all processes have offered a regular-V_t transistor offering an order of magnitude lower I_{off} at the expense of a 15% reduction in I_{dsat}. Some low-power processes provide a high-V_t transistor to reduce leakage by another order of magnitude. Most manufacturers use a separate implant mask to specify the threshold voltage, but Intel reduces manufacturing cost by using a slightly (~10%) longer channel length instead, which increases V_t on account of the short-channel effect [Rusu07].

Reported subthreshold slopes range from 85–100 mV/decade. DIBL coefficients range from 100–130 mV/V and tend to get larger with technology scaling.

Summary

CMOS process options and directions can greatly influence design decisions. Frequently, the combination of performance and cost possibilities in a new process can provide new product opportunities that were not available previously. Similarly, venerable processes can offer good opportunities with the right product.

One issue that has to be kept in mind is the ever-increasing cost of having a CMOS design fabricated in a leading-edge process. Mask cost for critical layers is in the vicinity of $100K per mask. A full mask set for a 65 nm process can exceed $1M in cost, and the price has been roughly doubling at each technology node. This in turn is reflected in the types of design and approaches to design that are employed for CMOS chips of the future. For instance, making a design programmable so that it can have a longer product life is a good first start. Chapter 14 covers these approaches in depth.

For more advanced reading on silicon processing, consult textbooks such as [Wolf00].

Exercises

3.1 A 248 nm UV step and scan machine costs $10M and can produce 80 300 mm diameter, 90 nm node wafers per hour. A 193 nm UV step and scan machine costs $40M and can process 20 300 mm diameter, 50 nm node wafers per hour. If the machines have a depreciation period of four years, what is the difference in the cost per chip for a chip that occupies 50 square mm at 90 nm resolution if the stepper is used 10 times per process run for the critical layers?

3.2 If the gate oxide thickness in a SiO_2-based structure is 2 nm, what would be the thickness of an HfO_2-based dielectric providing the same capacitance?

3.3 Explain the difference between a polycide and a salicide CMOS process. Which would be likely to have higher performance and why?

3.4 Draw the layout for a pMOS transistor in an n-well process that has active, p-select, n-select, polysilicon, contact, and metal1 masks. Include the well contact to V_{DD}.

3.5 What is the lowest resistance metal for interconnect? Why isn't it used?

3.6 Calculate the minimum contacted pitch as shown in Figure 3.39 for metal1 in terms of λ using the SUBM rules. Is there a wiring strategy that can reduce this pitch?

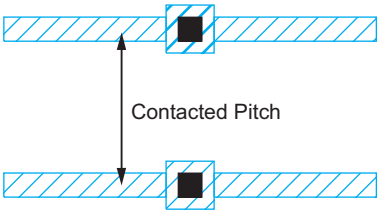


FIGURE 3.39 Contacted metal pitch

3.7 Using the SUBM rules, calculate the minimum uncontacted and contacted transistor pitch, as shown in Figure 3.40.

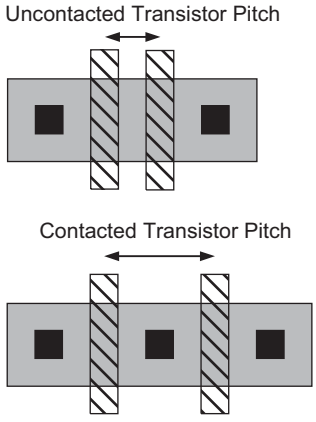


FIGURE 3.40 Uncontacted and contacted transistor pitch

3.8 Using Figure 3.41 and the SUBM design rules, calculate the minimum n to p pitch and the minimum inverter height with and without the poly contact to the gate (in). If an SOI process has 2 λ spacing between n and p diffusion, to what are the two pitches reduced?

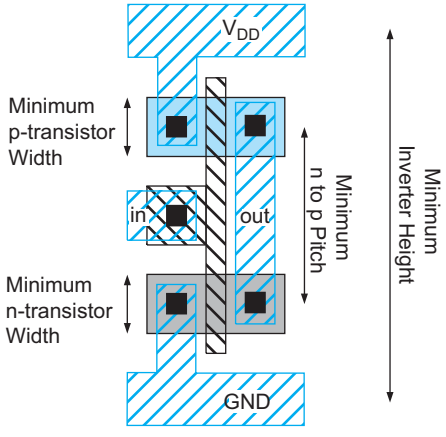


FIGURE 3.41 Minimum inverter height

3.9 Design a metal6 fuse ROM cell in a process where the minimum metal width is 0.5 μm and the maximum current density is 2 mA/μm. A fuse current of less than 10 mA is desired.